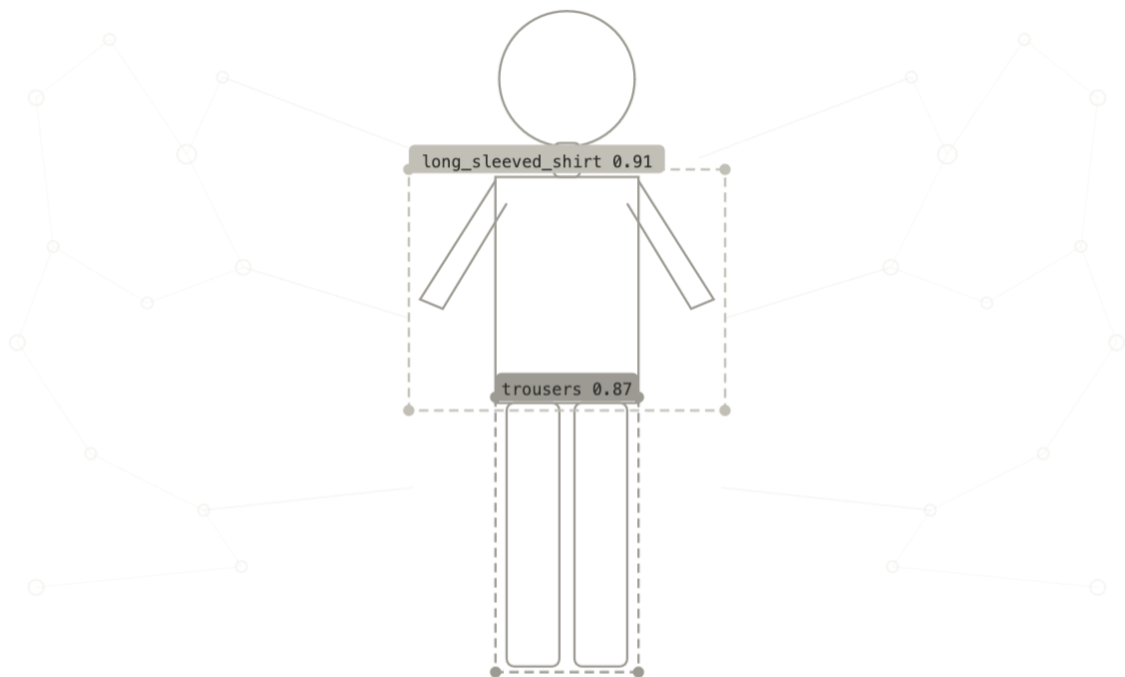


Dressed by the Algorithm

A Computer Vision and Machine Learning Approach
to Fashion Analysis and Consumer Behaviour on Instagram

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ABSTRACT

The fashion industry is continually expanding and evolving, and social media plays a significant role in shaping how consumers discover, engage with and express fashion. As a result, designers often turn to social media to gain fashion insight, as consumer-generated content on platforms such as Instagram has become a primary site of fashion communication. The need for automated and systematic data-driven frameworks for analyzing visual fashion signals and consumer behaviour has therefore grown. This study develops two complementary end-to-end frameworks leveraging computer vision and machine learning. The first framework applies a visual attribute extraction pipeline to 397 images, extracting dominant colour, pattern complexity and refined garment type using YOLOv8, K-means clustering, Laplacian variance analysis and CLIP zero-shot classification. The second applies a Random Forest classifier to 24,752 Instagram posts associated with fashion brands to classify consumer behaviour based on engagement and visual content features. The results demonstrate that visual fashion attributes show distinct patterns across garment types, colours and patterns, and that consumer behaviour differs systematically across brand categories. This study contributes a reproducible framework for fashion attribute extraction and consumer classification from social media data.

Keywords: Fashion; Information retrieval; Classification; Detection; Social Media; Machine learning; Consumer behaviour

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Thank you

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1 Introduction

The Digital Wardrobe

The modern-day society faces many diverse and fascinating challenges, and one of the most intense and important challenges is how to understand and shape the new technology revolution, known as the Fourth Industrial Revolution (Schwab, 2016). The changes are historic in terms of their size, speed, and scope, and the revolution entails nothing less than a transformation of humankind. It's defined by the fusion between the physical, digital, and biological worlds. This era is, unlike the other Industrial Revolutions, shaped by artificial intelligence, biotechnology, robotics, big data, and hyperconnectivity. The world has witnessed profound shifts across all industries, marked by the emergence of new business models, reshaping production and consumption. The fundamental shifts are not simply technological. On the societal front, a paradigm shift has significantly changed how societies work and communicate, as well as the way they express, inform and entertain themselves. One of the most visible reflections of this transformation can be seen in fashion and how people choose to express their identity (Jin & Shin, 2021).

The fashion industry is one of the many industries affected by the fourth industrial revolution and its technological transformation. AI, machine learning, big data, hyperconnectivity, and digital design tools are increasingly integrated into product development and planning processes. Whether it's 3D virtual prototyping to create real-life visualizations during the idea stage or retailers using generative AI as online shopping assistants powered to provide tailored recommendations and insights to users based on specific contexts (Jin & Shin 2021). The terms "creativity" and "innovation" are often seen as synonymous, and even the late Karl Lagerfeld used to say that "*the essence of fashion is changeability*" (Biehlmann, 2023). The fashion industry's adaptation to this technological shift is therefore deeply intertwined with innovation and digital transformation. The digital acceleration has created a hyperconnected ecosystem, where fashion is not only displayed but actively shaped and amplified through user-generated content and algorithmic recommendation systems. The revolution has created a shift in the rules of the power game in fashion (Kim, 2017).

A single TikTok video can make burgundy red the next big color within days. A viral Instagram post can rapidly transform niche aesthetics into globally recognized fashion movements. A Pinterest photo can push a forgotten silhouette back into relevance overnight. These posts can

explode across social media feeds at algorithmic speed before brands have time to react. This shift is the reason why fashion trends emerge and fade instantly, fueled by influencer culture and digital platforms, creating a “here today, gone tomorrow” environment.

This rapid change has fundamentally reshaped how the fashion industry operates. Brands are no longer the sole architects of consumer taste. Instead, brands must navigate a landscape where trends emerge organically from the bottom up, driven by everyday consumers on social media rather than by designers and runways. Brands are faced with a significant disruption because of the forces of media and technology. High-end luxury fashion brands once dominated absolute power as trendsetters, and their influence trickled down to lower-end fashion or mass fashion who followed the trends set by exclusive brands (Kim, 2017).

In the meantime, traditional print fashion magazines such as Vogue and Harper’s Bazaar were powerful and the main sources of information about upcoming fashion trends.

However, technology has yet again made a shift in the rules of the power game in fashion (Kim, 2017).

This change can be illustrated by one of cinema’s most memorable fashion moments:

“That sweater is not just blue. It’s Cerulean. And you’re also blithely unaware of the fact that in 2002, Oscar de la Renta did a collection of cerulean gowns. And then it filtered down through department stores and then trickled on down into some tragic Casual Corner where you, no doubt, fished it out of some clearance bin.”

- Miranda Priestly, *The Devil Wears Prada* (2006)

In 2006, Miranda Priestly delivered what would become one of cinema’s most memorable monologues in fashion. As she explained, a cerulean sweater did not simply appear by chance. Fashion trends were chosen by elite designers before filtering downward to the mass market. Fashion, in her view, moved in a largely top-down direction, shaped and controlled by a powerful few.

Twenty years later, *The Devil Wears Prada 2* (2026) presents the same Miranda Priestly fighting to preserve her print empire against what the film describes as the “tsunami of social media”. During an interview with ABC7, Meryl Streep was asked how Miranda Priestly would navigate Instagram today, responding: "She would have several assistants following people for her". The irony of this evolution is telling. Even the most powerful gatekeeper in fashion today

must pay attention to what consumers are posting on social media. This thesis examines how the manual observation of digital fashion signals can be systematized through data science methods.

Technology is changing every element of this power structure. In today's digital landscape, fashion trends no longer evolve gradually. They are no longer determined by luxury brands and traditional print magazines, whose influence has declined significantly in the digital era. Fashion trends emerge, spread, and disappear at an unprecedented speed. This change creates an opportunity but also a profound challenge for brands. This development raises an important question: *"What can social media data tell us about how consumers express and engage with fashion online?"*.

The relevance of this question lies in the growing disconnect between how fashion is consumed digitally and how brands understand that consumption. Traditional approaches to understanding consumer behavior have relied on sales data, surveys, and seasonal intuition. Methods that might struggle to capture dynamic, real-time signals generated by consumers on social media platforms such as Instagram. As a result, brands risk making decisions based on incomplete ideas of what consumers are wearing, sharing, and engaging online. In an industry already criticized for its sustainability challenges, a poor understanding of genuine consumer behavior is no longer just a business risk. It is an ethical one (Chen et al., 2024). Developing systematic frameworks for extracting and analyzing fashion signals from social media is therefore not only academically relevant, but also practically necessary.

At the center of this transformation lies data. Social media platforms generate massive amounts of multimodal data such as images, captions, hashtags, and engagement metrics. These platforms do more than display fashion; they actively shape it. They amplify aesthetics, reward visibility, and accelerate trend diffusion through algorithms. Hidden within this constant stream of digital expression are signals of emerging preferences. Colors gain momentum, silhouettes resurge, patterns quietly accumulate attention. The data reflect consumer preferences in real-time and represent a largely untapped resource for understanding consumer behavior.

The question is no longer whether social media influences fashion. It clearly does. The question is whether these digital traces can be systematically captured, analyzed, and transformed into actionable knowledge about consumer behavior. Given that the ultimate goal of technology development is solving industry challenges to better serve human needs (Schwab, 2016), this thesis argues that the vast stream of fashion-related content generated daily on social media

represents a largely untapped resource. A resource that with the right tools, can reveal meaningful patterns in how consumers express, engage with, and respond to fashion online.

Yet the sheer volume of this data presents its own challenge. Gartner defines dark data as the information assets organizations collect, process, and store during regular business activities, but generally fail to use for other purposes such as analytics, storing latent signals that go untapped (Hand, 2020). It is estimated that most organizations actively analyze as little as one percent of the data available to them. In the context of fashion, this means that vast volumes of social media signals are continuously generated but rarely subjected to systematic analysis. The opportunity lies not in the data itself, but in the capacity to transform it into structured, actionable knowledge.

The stated challenge is further complicated by a present and growing threat to data integrity. As AI-generated content proliferates across digital platforms, the boundary between authentic and synthetic consumer signals is becoming increasingly difficult to discern. Lautrup (2025) terms this phenomenon of the Digital Kessler Syndrome. The terminology draws on the space debris metaphor in which cascading collisions render an orbital environment unusable, and it is used to describe how the accumulation of synthetic data risks contaminating and diluting decades of authentic data collection. In context of this thesis and social media, the dynamic raises a methodological concern. If an increasing proportion of fashion content is AI-generated rather than reflecting genuine consumer expression, the validity of social media signals may erode over time. This concern points out the urgency of developing robust, systematic frameworks for extracting genuine fashion signals from social media data.

This thesis seeks to examine what consumers wear and how they engage with it on Instagram. Using two complementary datasets. One is a collection of street-style images from real users. The other is a metadata dataset of Instagram posts associated with fashion brands. In this thesis, computer vision and machine learning will be applied to extract visual fashion attributes and classify consumer behavior across brand categories. The result is a data-driven framework that aims to capture fashion from the ground and up. But in an era where style is shaped by scrolls, and where even Miranda Priestly needs assistant to monitor Instagram, this thesis asks a fundamental question: *“What can the data reveal about what people actually wear and how they express and engage with fashion online?”*. This study aims to contribute to the understanding of how fashion is expressed, consumed, and amplified on Instagram. For

decades, the fashion industry has operated on intuition, an experienced buyer's ability to sense what will sell, a designer's instinct for what consumers want. This study argues visual signals, engagement patterns, and the behavioural traces that consumers upload on social media encode preferences that can be systematically extracted and analyzed. This thesis proposes that computer vision and machine learning can surface patterns in data that are invisible to the naked eye but meaningful to the industry.

1.1 Research Question and Hypotheses

Research Questions

The following research questions are designed to guide the exploration of how social media data can be used to extract fashion attributes and classify consumer behavior through modern data science methodologies. The study draws on two complementary datasets: a collection of images analyzed using computer vision, and a metadata dataset of Instagram posts examined through machine learning classification.

The overarching research question guiding this study is:

How can computer vision and machine learning be applied to social media data to extract fashion attributes and classify consumer behavior across fashion brands categories on Instagram?

Given the broad scope of this research question, it is necessary to specify the core objectives that the analytical framework must accomplish. The framework must first extract meaningful visual fashion attributes from consumer-generated social media content using computer vision techniques. Furthermore, it must classify consumer behaviour across fashion brand categories based on engagement and visual content features. Finally, it must assess whether visual and engagement signals improves classification performance compared to single-modality approaches.

To explore these aspects systematically, three sub-questions have been defined to guide this study:

RQ 1: What fashion attributes can be automatically extracted from images using YOLOv8, and what characterizes the visual content of consumer-generated fashion images?

RQ 2: Can machine learning models classify Instagram content into fashion brand categories based on engagement and visual features, and what features drive this classification?

RQ 3: How do visual attributes from images and consumer engagement patterns on Instagram complement each other in understanding fashion behavior on social media?

Hypothesis Development

Building on the research questions, three hypotheses are proposed to guide the empirical analysis. These hypotheses reflect the study's focus on multimodal signals and the complementarity of visual and behavioral data.

Understanding fashion on social media requires identifying which types of signals contain meaningful information about consumer behaviour and visual expression. Social media platforms provide visual signals such as images, textual signals like captions and hashtags, and behavioral signals such as likes and comments. All of which may contain information about how consumers present and engage with different brands in fashion.

Previous research suggests that combining multiple data modalities often improves analytical performance. Jain and Singh (2025) use machine learning on Instagram and Twitter images, caption, hashtags and engagement metric to forecast emerging trends and find that multimodal models outperform unimodal alternatives in trend prediction tasks. Similarly, Social4Fashion (Balloni et al., 2024) demonstrates how automated visual extraction from social media can be used for fashion analysis, showing that computer vision techniques can reliably detect and classify garments from images. While these studies differ in scope and focus, both demonstrate the value of systematic, automated approaches to fashion analysis on social media.

Building on this literature, this study evaluates the contribution of visual and behavioural signals from social media to fashion attribute extraction and consumer behaviour classification. The following hypotheses are proposed:

H1: Visual Attribute Hypothesis

Visual fashion attributes extracted from images using automated computer vision techniques like YOLOv8 show distinct distributional patterns across garment types, colors, and patterns.

H2: Multimodal Superiority Hypothesis

A machine learning model combining visual and engagement features outperforms single modality models in classifying Instagram content into fashion brand categories

H3: Consumer Behavior Hypothesis

Consumer-generated fashion content displays distinct engagement patterns, and these patterns vary systematically across fashion brand categories (High Street, Designer, Small Couture, Mega Couture).

1.2 Thesis Structure

This section presents an overview of the thesis structure and the objectives of each section, aiming to guide the research process and facilitate the reader's understanding.

The purpose of the *Introduction* is to present the motivation of the thesis and research objectives. In addition to that, it is to provide a brief overview of the methodology employed for this analysis.

In the *Literature Review*, the existing scholarly literature on data-driven fashion analysis is critically examined across four dimensions. The chapter begins by situating the study within the broader theoretical and empirical debates surrounding traditional versus data-driven approaches, with particular attention to the role of digitalization and hyperconnectivity in reshaping the fashion industry. The review will then map the dominant methodologies and AI models employed in prior research, including supervised learning, deep learning architectures such as CNNs, object detection models and zero-shot classification framework. Subsequently, the chapter reviews the role of big data, tech and social media platforms as primary data sources and tools for fashion analysis, before identifying areas of consensus, contradiction, and omission in the existing literature. The chapter concludes by establishing the research gap that motivates the present study, namely the limited empirical evaluation of complementary two-pipeline frameworks combining computer vision and machine learning for visual fashion attribute extraction and consumer behaviour classification from social media data.

The *Theory* chapter introduces and discusses the machine learning and computer vision concepts and models used in this study.

In the *Methodology* chapter, the study's research design and approach will be presented, including the philosophical and methodological choice that underpin the study. The chapter is structured around two separate analytical pipelines, each applied to a distinct dataset. The first pipeline applies computer vision techniques to a collection of images to extract visual fashion attributes. The second pipeline applies machine learning classification to a metadata dataset of Instagram posts used to analyze consumer behaviour across fashion brands. The chapter provides a technical explanation of each pipeline and the methods used. Lastly, the chapter will identify limitations.

In the *Results* chapter, there will be a presentation of the findings given by the model outputs and evaluation results, including relevant metrics as well as examples of output from the methodology when applied to the visual attribute extraction and consumer behaviour classification pipelines.

The *Discussion* chapter's purpose is to answer the research questions and delve into the interpretation and implications of the results. In this chapter, the study will furthermore compare the results to existing research. Lastly, the chapter will provide a reference for future research in the area.

The *Conclusion* will provide a summary of the thesis research objectives and highlight the main findings from the study.

2 Literature Review

Who Wore it Best? - The Archive

This chapter critically reviews existing scholarly literature relevant to the study. The purpose of the literature review is fourfold: (i) to situate the study within existing theoretical and empirical debates, (ii) to identify dominant concepts, models, and methodologies employed by previous researchers, (iii) to critically examine areas of consensus, contradiction, and omission, and (iv) to establish the research gap that the present study seeks to address. The review is thematically organized and aligned with the study objectives and research questions, ensuring coherence between prior scholarship and the proposed inquiry.

2.1 Theoretical and Empirical debates

The scholarly discourse regarding fashion analysis is situated at the intersection of traditional artistic intuition and modern data-driven methodologies. The fashion industry operates within a broader context of the rapid digital transformation and the fourth industrial revolution's technologies (Schwab, 2016; Oesterreich, 2020). The integration of big data analytics, automation, and digital platforms has reconfigured supply chains, consumer engagement strategies, and forecasting mechanisms (Balloni, 2024). Within this environment, hyperconnectivity is defined by continuous digital interaction. Real-time and constant content exchange has altered how fashion trends emerge and disappear.

Historically, fashion has been defined as a human-centric process where designers and stylists analyze culture, societal attitudes, and current events to predict style movements (DuBreuil, 2020; Grammenos, 2015). This traditional paradigm relies heavily on the “gut feeling” and artistic vision of experts, which critics often dismiss as “opinionated guesswork” involving high business risk. (Tehrani & Ahrens, 2016; Fox, Graul, & Peng, 2018). However, modern digital ecosystems enable trend formation driven by user-generated content and influencer culture (DuBreuil, 2020; Silvia, 2019). Leading to an accelerated fashion trend cycle, increasing volatility and reducing predictability (Savolainen, 2023; Giri, 2022).

Scholars argue that fashion brands increasingly rely on digital platforms and analytics to respond to volatile consumer preferences and the shortened product lifecycles (Balloni et al., 2024). This transformation has shifted fashion methods from intuition-based approaches to data-supported models capable of identifying emerging signals earlier in the trend cycle (Kontrimiene, 2017). This shift is often analyzed through the lens of management fashion theory, which examines how the fourth industrial revolution emerges as solutions for business problems such as intense global competition and the need for sustainability. In this context, fashion has evolved from a seasonal, runway-focused activity to a real-time, continuous process driven by digital globalization.

The broader context of digitalization and hyperconnectivity helps explain the implications and opportunities of leveraging data science methods in fashion analysis. The abundance of real-time social media data presents new possibilities for understanding consumer behaviour and visual fashion signals, while challenging traditional business frameworks. However, while technological methods dominate the field, the theoretical integration between online engagement patterns and actual consumer behaviour remains theoretically underspecified and weak.

2.2 Dominant Methodologies and AI Models

Researchers have increasingly employed artificial intelligence and machine learning to automate and enhance fashion analysis and its accuracy. Supervised learning techniques are used to predict consumer demand and product popularity based on historical data patterns (Au, 2008). More advanced methodologies incorporate time-series modeling to capture the dynamic nature of fashion cycles (Kim, 2017).

More recent research adopts deep learning architectures like Convolutional Neural Networks (CNNs) to enable automated feature extraction like visual features from large-scale images dataset, and it improves classification and style detection accuracy (Jain, 2025). CNNs have become a dominant method for visual fashion analysis due to its ability to learn hierarchical representations of visual content directly from raw pixel data.

Social4Fashion demonstrates the significant methodological development in computer vision, by using models to not only classify what is present in an image but also locate and isolate individual items through bounding boxes. This was done by applying deep learning to filter fashion content, detect specific garments and retrieve dominant colours through clustering. A pipeline structurally similar to the one developed in this study.

More recently, vision-language models such as CLIP (Radford et al., 2021) have enabled zero-shot classification, by matching images to text descriptions without task-specific training data. For consumer behaviour analysis, methods like Random Forest have been adopted for their interpretability and ability to produce feature importance ranking that helps explain what drives the classification (Breiman, 2001)

Additionally, hybrid methods that combine deep learning and traditional statistical models demonstrate an improved predictive performance compared to single-modality frameworks, especially in environments characterized by rapid trend shifts (Jain & Singh, 2025). However, there are concerns about model interpretability and overfitting that remain prevalent in the literature (Chen, 2024).

2.3 The Role of Big Data and Social Media Sources

The literature identifies a significant consensus that social media and search engine data are superior to traditional e-commerce data for trend discovery (Au et al., 2008; Silvia et al., 2019). Platforms like Instagram, TikTok, and Pinterest provide a real-time stream of information where digital influencers now dictate trends more effectively than traditional brands (Balloni et al., 2024). Social4Fashion, an intelligent expert system for analyzing fashion trends from social media, has been developed to scrape digital platforms, leveraging deep learning to filter fashion content, detect specific garments like handbags and retrieve dominant colors through clustering. Furthermore, Google Trends has been established as a leading indicator for purchase decisions and brand engagement, even though research into its application for fashion remains relatively unexplored (Silvia et al., 2019).

A central methodological challenge in data-driven fashion analysis is the absence of direct measures of trend adoption or commercial success. Sales data is proprietary, and runway

validation operates on timescales incompatible with real-time social media dynamics. As a result, researchers have increasingly relied on proxy labels derived from observable digital signals to operationalize popularity. Engagement metrics such as likes and comments have been shown to correlate with consumer purchasing intent and trend diffusion, as exposure to high-engagement content has been found to influence shopping habits (Jain & Singh, 2025; Silvia et al., 2019). These signals provide a foundation for understanding how consumers interact with fashion content on social media

Similarly, post frequency over time has been employed as an indicator of organic trend emergence, capturing the rate at which a given attribute accumulates visibility across a platform. Taken together, engagement and frequency constitute a composite signal that reflects both audience receptivity and the breadth of trend diffusion, offering a more robust proxy for popularity than a single measure alone (Balloni et al., 2024).

Beyond Academic Research

The appetite for data-driven fashion analysis is evident in practice and on social media. Namely, the Instagram account @databutmakeitfashion, founded by data scientist Madé Lapuerta in 2019. Data, but make it fashion has nearly 500,000 followers on Instagram by applying data analysis to fashion trends. This demonstrates a significant public interest in the systematic analysis of fashion signals from social media, runways etc. Similarly, the account @allthingsfashiontech documents the intersection of technology and fashion, this reflects a growing interest within the field. The gap between practitioner interest and academic framework motivates the present study.

2.4 Areas of Consensus, Contradiction, and Omission

Throughout the literature, there is a broad agreement that AI is highly effective at extracting and classifying colours and patterns from fashion imagery. This consensus stems from these attributes being more straightforward and predictable (DuBreuil, 2020). Studies by Kin-Fan Au (2008), Bhuiyan Uddin (2020), Choi T (2014) and Giri C (2024) argue that fashion demand is increasingly volatile and time-dependent, necessitating automated support systems to maintain high inventory fill rates. While these studies focus primarily on demand forecasting,

they demonstrate a broader consensus that automated attribute analysis is essential for understanding consumer behaviour in a volatile fashion market.

A major debate exists regarding the predictability and classification of complex design details such as necklines and hem lengths. Design details are particularly difficult to forecast, as they are shaped by cultural and emotional factors that resist quantification. Some studies have shown that human-based forecasting tools like WGSN and big-data tools like EDITED are least similar in this area, as consumer desire for unique design is shaped by unpredictable cultural and emotional factors that automated systems struggle to process (DuBreuil, 2020). Additionally, the effectiveness of CNNs remains contested, with some researchers noting issues with image positioning affecting results (Savolainen, 2023). This is relevant to this study, as garment detection and attribute extraction face similar challenges when garments are photographed from non-standard angles.

The literature reveals a notable lack of research on the social and ethical implications of AI adoption in fashion, in contrast to the heavy focus on technical and economic gains (Oesterreich, 2020). There is additionally a significant omission regarding how these automated analytical tools can benefit sustainable fashion brands, which face unique challenges in creating timeless designs and managing small-scale supply chains (Savolainen, 2023).

2.5 Establishing the Research Gap

Despite the growing use of AI models and computer vision in fashion analysis, several critical research gaps remain in the literature. A notable gap exists in the empirical evaluation of big data tools versus traditional human-centric analysis methods for creative activities like style prediction. This research gap exists due to the literature disproportionately emphasizing merchandising, trend forecasting and inventory optimization (DuBreuil, 2020; Savolainen, 2023). Leaving the systematic analysis of what consumers actually wear and how they engage with fashion on social media relatively underexplored.

A geographical gap is also evident, as the existing research is heavily concentrated in North America, Europe and East Asia leaving fashion markets in other regions largely unrepresented and limiting the generalizability of current findings (Chen S, 2025; Kim 2017). The

geographical concentration of fashion research reveals a promising but also largely unexplored research area, a point underscored by Murzyn-Kupisz (2025) who argues that geographic research has significant potential to add a more nuanced socio-economic and spatial perspective to fashion studies. Research confirms that the scholarly output is strongly concentrated in economically advanced and industrialized nations, resulting in datasets, platforms and consumer behaviors that disproportionately reflect Western and East Asian fashion contexts.

A further gap exists in the literature regarding frameworks capable of capturing emotionally and culturally driven trend movements and consumer behaviour, where fashion is shaped by unpredictable cultural moments rather than gradual aesthetic evolution. The 2023 *Barbie* film offers a compelling illustration of the commercial significance of this gap. The release of the 2023 *Barbie* film became a “social phenomenon” and orchestrated a massive cultural and commercial re-evaluation of pink, transforming it from a shade associated with childhood into a symbol of empowerment and strength. The trend’s momentum was amplified through social media, with users playing a core role in spreading the “Barbiecore” aesthetic on TikTok and Instagram. The film had an equally significant commercial impact, with sales of pink items increasing between Q2 2022 and Q2 2023. According to Bitran (2023) the sales of pink skirts increased by 66.8% and pink matching sets by 73.5% between Q2 2022 and Q3 2023.

Notably, the industry forecaster WGSN had identified early indicators of Barbie Pink gaining traction across different regions, demonstrating that data-driven tools have a capacity to anticipate the trend ahead of the cultural moment. However, the commercial performance of hyper-pink products fell below expectations, which suggests that current models struggle to accurately capture the magnitude of culturally driven trends and consumer behaviour. A comparable example is the neon green aesthetic of Charli XCX’s *Brat* album in 2024, which generated a singular cultural moment. These cases illustrate that culturally triggered trends and consumer behaviour operates through mechanisms such as emotional resonance, celebrity association, and viral diffusion. In addition, the cases of *Brat* and *Barbie* illustrate how trends can emerge and be shaped by unpredictable cultural moments that significantly impact consumer shopping trends, which existing quantitative frameworks are not designed to capture. Developing robust frameworks for this type of analysis represents a significant opportunity for future research.

Additionally, the potential for AI to support sustainable fashion through better understanding of consumer behaviour, rather than reinforcing the fast fashion bias present in current models, remains an underexplored area with significant implications for the industry's future (Savolainen, 2023).

Although these research gaps remain noteworthy and meaningful opportunities for future research, addressing them lies outside the objectives of the current study.

2.6 Contribution of the Study

Balloni et al. (2024) evaluate their Social4Fashion system primarily on classification and detection accuracy, demonstrating that automated garment detection and colour extraction from social media is technically feasible. However, their study does not examine how detected visual attributes relate to consumer engagement patterns, nor does Social4Fashion analyze how consumers across different brand use social media to present and engage with fashion. Kim et al. (2020) collect and analyze Instagram posts associated with fashion influencers, using engagement metrics as metadata rather than as analytical targets. Neither study combines visual attribute extraction with consumer behaviour classification across brand categories.

Existing studies tend to rely on heterogeneous indicators, including sales volume and consumer demand (Au et al., 2008; Choi et al., 2014), search indices as leading indicators of purchase intent (Silvia et al., 2019), trend co-occurrence frequency derived from social media (Ma et al., 2020), or combined engagement metrics such as likes and comments (Jain & Singh, 2025). However, these approaches often lack systematic justification and are difficult to compare across studies. The studies tend to address visual or the behavioural dimension of fashion on social media in isolation. Visual studies focus on garment detection and attribute extraction (Balloni et al., 2024; Ge et al., 2019), while behavioural studies focus on engagement metrics and brand performance (Ha et al., 2017; Jain & Singh, 2025). The relationships between what consumers visually wear and how they engage with fashion brands on social media remains underexplored as a combined analytical framework.

The present study addresses this gap by constructing three contributions. First, this study develops a visual attribute extraction pipeline that combines YOLOv8 garment detection, K-means colour extraction, Laplacian variance pattern detection, and CLIP zero-shot garment type refinement into a reproducible framework for analyzing images. Second, it applies machine learning classification to Instagram metadata to examine consumer behaviour across four fashion brand categories. Third, the study introduces a hierarchical taxonomy of 13 broad parent categories and 53 fine-grained sub-categories, which serves as a methodological contribution that future research can build upon and extend.

While the geographical gap and the challenge of culturally triggered trend movements identified in section 2.5 remain outside the scope of this study, the proposed framework establishes a foundation that future research can build upon by extending to larger and more recent datasets, underrepresented fashion markets or incorporating cultural event signals as additional analytical features.

In summary this study provides a reproducible and theoretically grounded framework for visual fashion attribute extraction and consumer behaviour classification derived from social media data. The approach responds to the methodological gap identified in section 2.5, namely the disproportionate emphasis on inventory optimization and forecasting, and the absence of empirically evaluated frameworks for analyzing what consumers actually wear and how they engage with fashion brands on Instagram.

This framework provides a direct foundation for the methodological structure of this study. By combining computer vision techniques for visual attribute extraction with machine learning classification for consumer behaviour analysis, it becomes possible to systematically examine fashion signals across two complementary datasets, evaluate model performance, and compare findings across brand categories within a unified, data-driven framework. The following section details the theoretical concepts underpinning this framework.

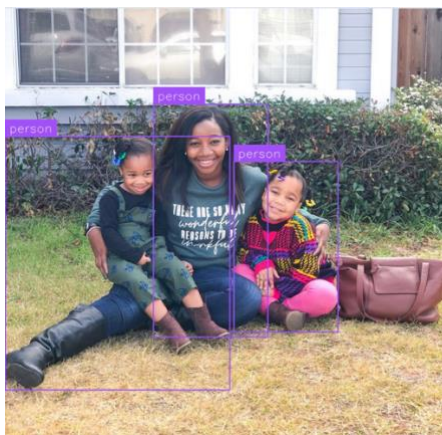
3 Theoretical Framework

Learning the Language of Fashion

This section outlines the central concepts and techniques in machine learning (ML) and computer vision that underpin the modelling framework used in this study. Although both fields cover a broad spectrum of methods and applications, this introduction is restricted to those most relevant for establishing the theoretical foundation and supporting the methodological choices made in this research as described in the analytical pipelines in Chapter 4. Namely the extraction of visual and textual fashion attributes from social media data. Each section provides a conceptual foundation for the methodological choices made in the study.

3.1 Object Detection – YOLOv8

Object detection is in this study carried out using YOLOv8 (You Only Look Once), a state-of-the-art, real time object detection and image segmentation model developed by Ultralytics. The detection architecture is fast, accurate and well-documented in the literature, and because the model scans the entire image in a single pass, it is efficient and returns precise bounding boxes around each detected object. Figure 3.1 shows example detections.



3 objects detected

person



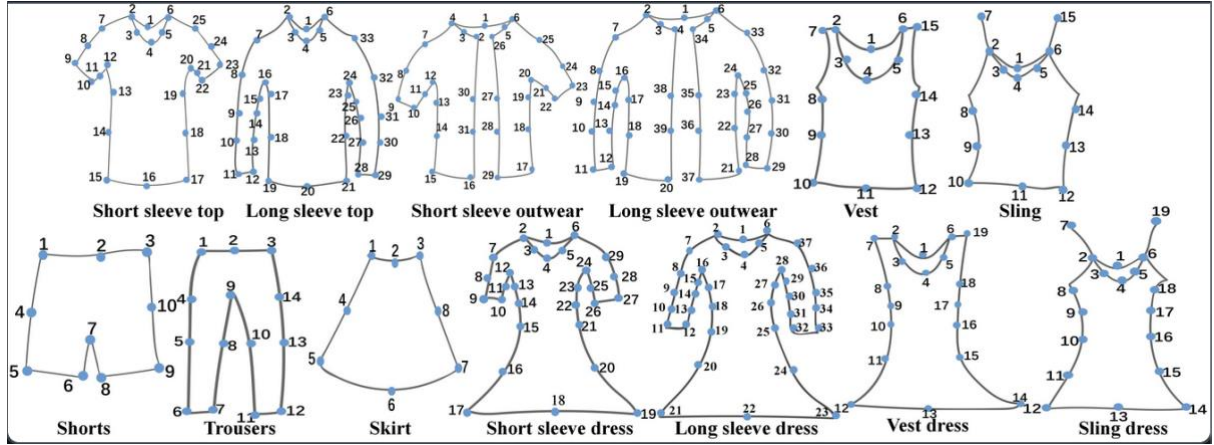
2 objects detected

refrigerator

bottle

“Figure 3.1: YOLOv8 detection examples. Left: Three persons detected illustrated with bounding boxes in a street-style image. Right: non-fashion content detected and rejected by fashion filter. Source Roboflow, illustrated using COCO weights for visualization purposes. The production pipeline uses DeepFashion2 weights”.

YOLOv8 models are trained on the COCO dataset, which recognizes only general objects such as “person” and “bottle” and does not recognize specific garment types. For this reason, the model is used together with DeepFashion2 weights, which give it fashion-specific knowledge across 13 garment categories such as long sleeve top, trousers, skirt and dress, as illustrated in figure 3.2. Without DeepFashion2 weights, the model would not be able to distinguish between a jacket and a dress.



“Figure 3.2: The 13 garments categories defined in DeepFashion2 with their corresponding landmark annotations. Source: Ge et al. (2019)

DeepFashion2 is a large-scale fashion dataset developed by Ge et al. (2019). It achieves a mean Average Precision of 0.849 at IoU 0.50 on the DeepFashion2 realistic clothes split, and it recognizes 13 garments categories at varying sleeve and length. Although DeepFashion2 is a segmentation model, only its bounding box output is used in this study. Figure 3.3 illustrates DeepFashion2 detection applied to multi-person scene. Together, YOLOv8 and DeepFashion2 enable the pipeline to locate garments in images but also assign them to meaningful fashion categories.



“Figure 3.3: DeepFashion2 garment detection applied to a multi-person scene, showing bounding boxes and category labels across multiple garments simultaneously. Source: Ge et al. (2019)”

3.2 Zero-Shot Classification

Zero-shot classification is provided in this study by CLIP (Contrastive Language–Image Pre-training), a model developed by OpenAI and trained on 400 million image–text pairs. This training gives CLIP the ability to match images to text descriptions without requiring any additional training or labelled fashion data, which is what allows it to assign garments to categories it was not explicitly trained on.

Two design choices govern the application of CLIP in this study. The prompt template used to present candidate labels to CLIP's text encoder, and the confidence threshold below which a prediction is marked as uncertain.

Following Radford et al. (2021), candidate labels are not passed to CLIP as bare nouns but as templated natural-language phrases, since CLIP was trained on image-caption pairs rather than isolated words. This study extends the standard single-template approach to a prompt ensemble: each candidate label is presented under four templates: "a photo of a {label}", "a photo of someone wearing a {label}", "a fashion photograph of a {label}", and "a close-up of a {label}". Logits are computed independently for each template and averaged before the final softmax, producing a more stable probability distribution that reduces sensitivity to the wording of any individual prompt.

The confidence threshold is implemented as a per-parent calibration rather than a single dataset-wide scalar. The motivation is mathematical. Softmax probability is distributed across the candidates within a parent, so a parent with many sub-labels mechanically produces lower top-1 probabilities even when the model is functioning correctly. A uniform threshold of 0.4, appropriate for a parent with two or three candidates, would over-abstain on parents with six or seven candidates.

Thresholds are therefore calibrated so that each parent's bar sits above the uniform-probability floor of $1/n_{\text{candidates}}$, but high enough to filter genuinely ambiguous predictions. Parents with rich vocabularies (long-sleeve top, long-sleeve outwear, trousers) use a threshold of 0.25; parents with medium vocabularies (short-sleeve top, skirt, shorts) use 0.30; all other parents use the default of 0.40.

3.3 Colour Clustering K-means

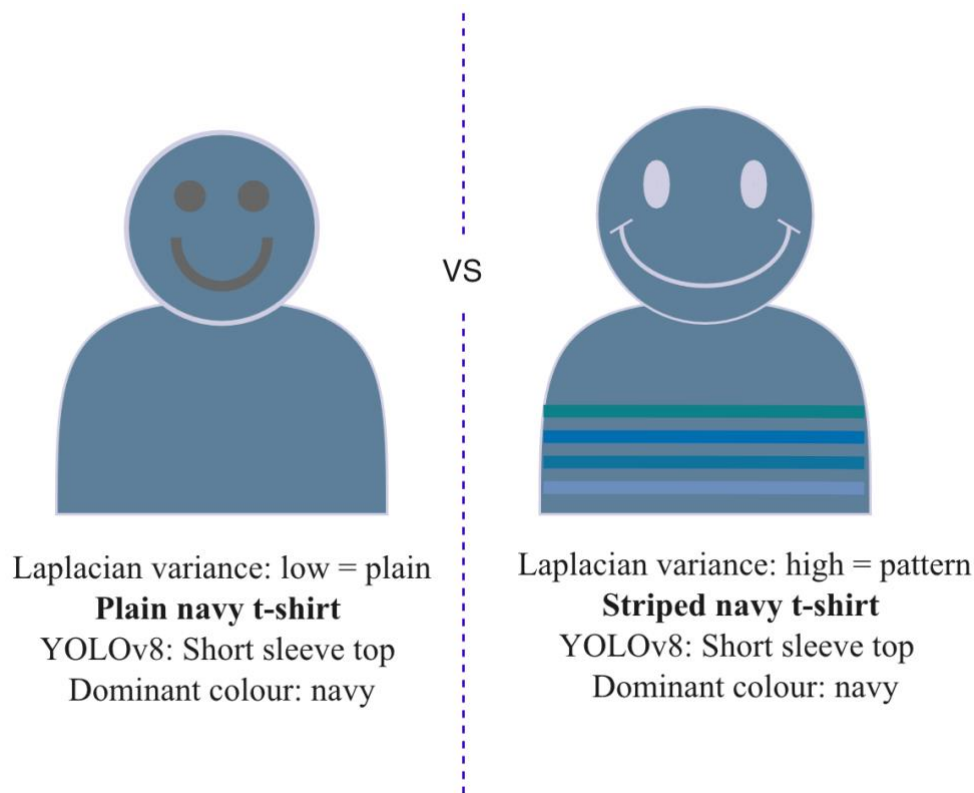
Dominant colour is extracted using K-means clustering (MacQueen, 1967), which groups the pixels of a garment into a small set of colour clusters, with the largest cluster by pixel count taken as the dominant colour.

Clustering is performed in the CIELAB colour space rather than RGB. The reason is that Euclidean distance in CIELAB space approximates human perceptual colour difference, producing dominant-colour assignments that better match visual judgement than equivalent operations in RGB. Once the dominant cluster has been identified in CIELAB space, it is converted back to RGB for storage and reporting, as RGB values are more universally interpretable for downstream analysis.

3.4 Laplacian Variance

Pattern complexity is measured using Laplacian variance analysis, first introduced by Pech-Pacheco et al. (2000) as a measure of image sharpness. The Laplacian operator detects edges in an image by measuring how sharply pixel values change between neighbouring areas. A plain, single-coloured garment produces very few edges and therefore a low Laplacian variance, whereas a patterned garment such as a floral dress or a striped shirt produces many edges and therefore a high variance. The variance of the Laplacian is thus used as a scalar

measure of pattern complexity (Figure 3.4). Because pattern complexity is a property of intensity rather than colour, the image is converted to greyscale before the Laplacian is applied.



“Figure 3.4: Two garments classified identically by YOLOv8 (short sleeve top, dominant colour: navy) but distinguished by pattern complexity. Own illustration.”

3.5 Random Forest Classification

Consumer behaviour classification is carried out using a Random Forest classifier (Breiman, 2001). Random Forest constructs a large number of decision trees during training, where each tree is built on a random subset of the data and a random subset of the features. The final prediction is obtained by aggregating the votes of all individual trees. This reduces the tendency of single decision trees to overfit and produces a more robust and stable classifier.

Random Forest is well suited to this study for several reasons. It handles mixed feature types and varying ranges well, which fits the heterogeneous feature set used here; it provides a built-in measure of feature importance, allowing the analysis to identify which features drive the

classification; and it performs reliably on tabular data of this scale while remaining robust to irrelevant or correlated features.

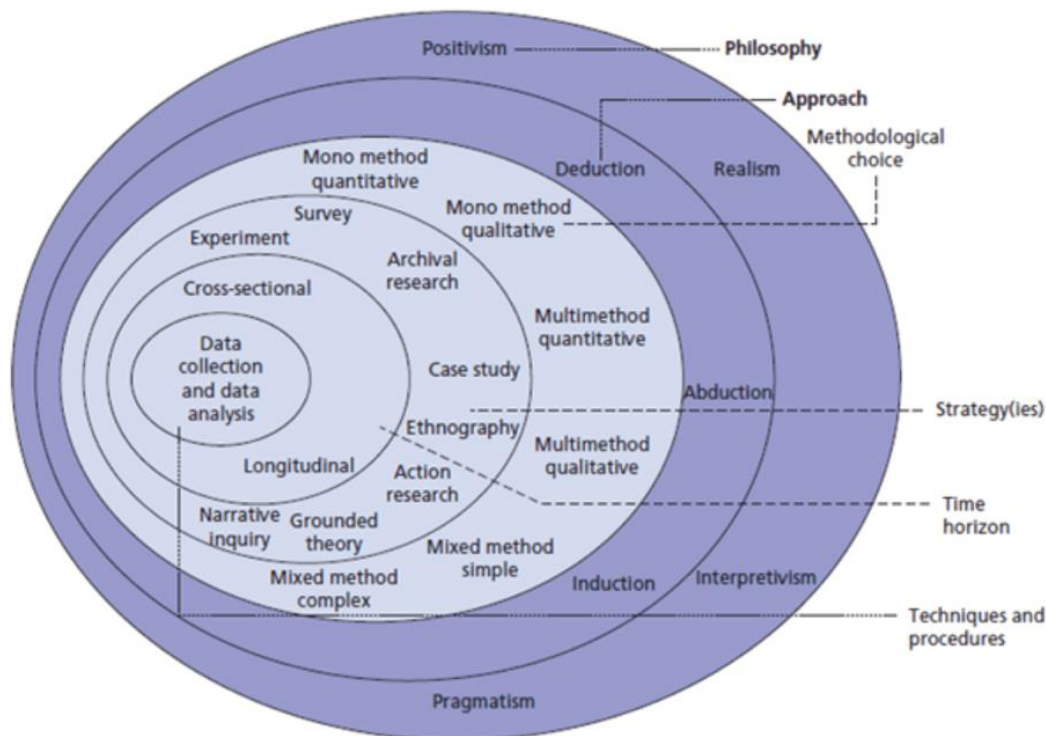
4 Methodology

Behind the seams: Stitching the Pipeline

This chapter presents the methodology used to address the research questions and details the techniques utilized for data collection, processing, modelling and evaluation. The subsequent sections aim to elaborate on the selected research philosophy, approach and design that guide this study and answer the research questions.

The methodological framework of this study, “Dressed by the Algorithm”, is structured around Saunders, Lewis and Thornhills’s (2019) Research Onion, illustrated in Figure 4.1. The figure organizes the research design across six layers, each layer of the research process is based on assumptions about the sources and the nature of knowledge. Moving from the outermost philosophical assumptions and slowly peeling inward towards the specific techniques and procedures of data collection and analysis. The following sections of the Methodology chapter peel each layer of the onion to position the study within its philosophical, methodological and technical context.

Due to the interdisciplinary nature of this study, which combines computer vision and machine learning techniques with the analysis of consumer behaviour on social media, the methodological framework draws on data science and social science. The Research Onion framework is adopted as a structuring device to make the philosophical and methodological choices explicit, which is a common practice in social science and business research.



“Figure 4.1: The Research Onion. Source: Saunders et al. (2019)”

4.1 Research Philosophy

Research philosophy is a critical aspect of some research projects. The term refers to the set of underlying beliefs and assumptions about *how* knowledge is developed, *what* constitutes valid knowledge, and *how* research should be conducted. The three fundamental components address, in Saunders definition on research philosophy, is known as ontology which is the nature of reality, epistemology as in the nature of knowledge and methodology the approach to inquiry.

Research philosophy provides a framework for understanding the nature of reality, knowledge and the methods used to acquire that knowledge (Saunders et al., 2009). In the context of this study’s research questions and thesis project, identifying and discussing the research philosophy is essential to ensure the coherence and validity of the chosen approach.

Saunders et al. (2019) outlines four main research philosophies in their research onion model: positivism, realism, interpretivism, and pragmatism. As illustrated in the first onion layer in figure 4.1. Positivism is a research philosophy that emphasizes the use of objective, quantitative methods to study social phenomena and aims to uncover generalizable laws. Realism, on the other hand, assumes that reality exists independently of human perceptions, and researchers seek to uncover reality with appropriate methods. Interpretivism focuses on understanding the subjective meanings and experiences of individuals through qualitative research methods. Pragmatism is centered around practical solutions and allows for the integration of both quantitative and qualitative methods to address the research objectives.

This study adopts a positivist research philosophy given that a social phenomena can be studied objectively through systematic observation and measurement. The following questions define the philosophical position of “Dressed by the Algorithm”.

Ontology – The Nature of Reality

This study is centered around a realist ontological position, assuming fashion attributes like colour, pattern, and garment type exist as objective and measurable properties. The assumption is defended by how the properties can be observed in images independently of the researcher. A purple skirt is purple regardless of who observes it, and a floral t-shirt is visually distinct from a mini skirt in a way that can be systematically detected and measured. The position uses automated computer vision methods to extract fashion attributes from social media images.

Epistemology – The Nature of Knowledge

The nature of knowledge in this study adopts a positivist epistemological position, arguing that valid knowledge about consumer fashion behavior can be derived through systematic, objective data. Instead of relying on qualitative interpretations Furthermore by classifying engagements patterns from metadata, and with digital signals such as likes, comments, and visual features are treated as valid proxies for consumer behaviour and preference.

Methodology – The Approach to Inquiry

The approach uses a deductive methodological approach, by using three theoretically grounded hypotheses and testing them empirically against two complementary datasets. Positivist philosophy favours hypothesis-driven inquiry and reproducible, quantitative methods. The

framework combines computer vision techniques with machine learning classification for the analysis which enables a systematic and comparable outputs.

4.2 Research Approach

The research approach conducts a roadmap for the implementation of a study. It describes the logical direction by outlining the methods and techniques that the research follows in order to answer the research question or test hypotheses. In line with the positivist research philosophy outlined in section 4.1, the study adopts a deductive research approach, focusing on the systematic application of computer vision and machine learning techniques to extract fashion attributes and classify consumer behaviour on Instagram. This approach makes it possible to test whether the patterns suggested by existing research actually appear in the data, in the way that is structured, measurable and can be replicated by others.

The three research questions that guide this study are:

RQ 1: What fashion attributes can be automatically extracted from images using YOLOv8, and what characterizes the visual content of consumer-generated fashion images?

RQ 2: Can machine learning models classify Instagram content into fashion brand categories based on engagement and visual features, and what features drive this classification?

RQ 3: How do visual attributes from images and consumer engagement patterns on Instagram complement each other in understanding fashion behavior on social media?

To address these questions, the study uses a multi-method quantitative approach by combining two Data Science methods for consumer behaviour analysis. This strategy enables a systematic and comparable examination of fashion-related signals across two complementary datasets, facilitating the empirical evaluation of the proposed hypotheses.

The choice of a deductive research approach is informed by the nature of the research questions, which seek to test established theoretical propositions about the value of visual and behavioural signals in fashion contexts. Deductive research is particularly well-suited for studies that draw on an existing body of literature to formulate hypotheses prior to data collection and analysis (Saunders et al., 2009). The prior research on data-driven fashion analysis, computer vision,

and social media engagement provides a strong theoretical foundation from which the hypotheses of this study are derived.

This study proceeds in two analytical stages. The first stage applies computer vision to a collection of 397 random images, employing YOLOv8 for garment detection, K-means clustering for colour extraction, Laplacian variance analysis for detection, and CLIP zero-shot classification for garment type refinement. The second analytical stage applies machine learning classification to a metadata dataset of Instagram posts, using a Random Forest classifier to predict fashion brand categories based on engagement metrics and visual content feature. The results from the stages are subsequently brought together in the discussion chapter in relation to the overarching research questions.

In summary, the deductive research approach provides a structured and theoretically grounded framework for examining how computer vision and machine learning can be applied to social media data to extract fashion attributes and classify consumer behaviour across fashion brand categories on Instagram, and where their relationship to the overarching research questions is examined.

4.3 Research Design

The research design serves as a blueprint for the study, outlining the process through which the study addresses the research questions and accomplishes the study's objectives. In the context of this research, three aspects of research design are relevant: the methodology choice, the research strategy, and the time horizon.

Methodology choice

The methodology choice adopts a multi-method quantitative design, by combining two distinct quantitative methods. This study does not provide any qualitative component like interviews, surveys or subjective interpretation of content. "Dressed by the Algorithm" relies entirely on measurable outputs from computer vision and machine learning models. The given choice aligns with the positivist philosophy and deductive approach described in section 4.1 and 4.2, it furthermore reflects the nature of data available. The datasets in this study is used for quantitative analysis, and the research questions calls for measurable and comparable results.

Research strategy

The research strategy describes how the study approaches its data and analysis. In the case of this study an archival research strategy is used, meaning it draws on existing data rather than collecting new data through experiments, surveys, or fieldwork. Neither of the datasets was created for this thesis specifically.

Time horizon

This section refers to the time period of the study meaning whether it looks at a single point in time or tracks changes over a period. The time horizon for this project takes a cross-sectional approach, by having datasets that represents snapshots of the point in time rather than a continuous stream of data over time. The images used were collected within a defined period, and the metadata reflects posts from a specific window of time. This means that the results describe what was present in the data at the time, and it does not capture how fashion attributes or engagements patterns change week by week or month by month. This is an important limitation that is addressed in the Limitations subsections of Sections 4.5 and 4.7.

Data Collection and analysis

In this section, the data understanding stage of the study is outlined. More specifically, the elaboration on the sources, methods, and considerations involved in acquiring suitable datasets for the study. This thesis draws on two existing datasets rather than collecting primary data. The image dataset consists of 397 photographs sourced from Seungbae Kim's Instagram research dataset.¹ The metadata dataset was obtained from Harvard Dataverse and consists of 24,752 Instagram posts associated with ten fashion brands.²

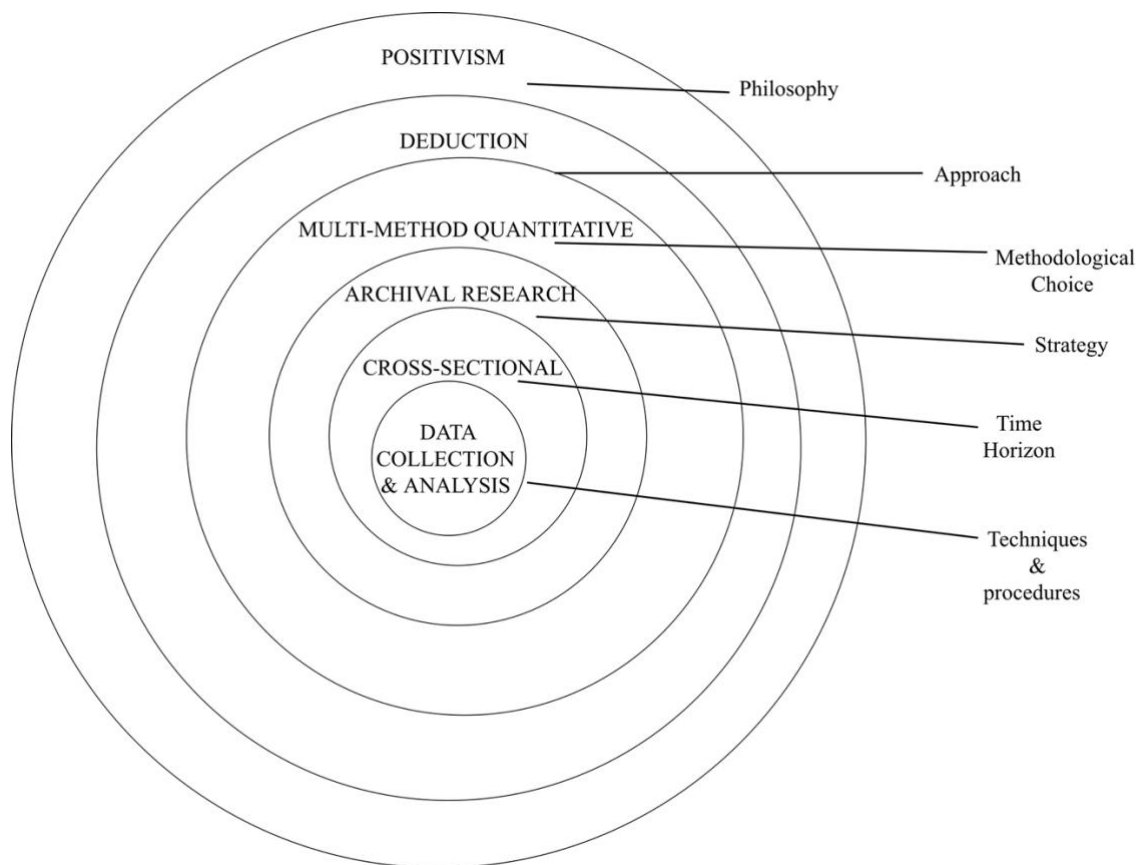
The datasets are described in detail in sections 4.4 and 4.5, whereas the data analysis is carried out through two separate pipelines described in section 4.5 and 4.7.

Summary of methodological choices

Before proceeding to the datasets and analytical pipelines, the methodological choices outline in sections 4.1, 4.2, and 4.3 are briefly summarized below. The methodology chapter continues in section 4.4.

¹ <https://sites.google.com/site/sbkimcv/dataset/instagram-influencer-dataset>

² <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/K7AW6F>



“Figure 4.2: The Research Onion adapted to the present study. Source: Saunders et al. (2019)”

In summary, this study uses the methodological framework around Saunders Research Onion, illustrated in figure 4.2. In the first and outermost philosophical layer of the onion this study is grounded in a positivist research philosophy, assuming that fashion attributes and consumer behaviour patterns are objectively observable and measurable. By slowly peeling inwards towards the next layer it is revealed that the study adopts a deductive approach, whereby three theoretically grounded hypotheses are tested against empirical data. These two outer layers center the fundamental beliefs about what knowledge is and how it is generated. The focus is on the philosophical context.

In addition, the study applies a multi-method quantitative design by drawing exclusively on measurable outputs from computer vision and machine learning models. The research strategy is archival, relying on two existing datasets rather than collecting primary data. These two middle layers are based on the practical decision about how the study is structured and the data used. This delivers the methodological context

The time horizon is cross-sectional instead of longitudinal observation. In the innermost layer is data collection and analysis which are carried out through two separate analytical pipelines which are brought together in the discussion chapter. These two inner layers derives from the specific methods and tools used to collect and analyze data which gives us the technical context. These methodological choices are visualized in Figure 4.2, which represents an adapted version of Saunders et al's (2019). Research Onion tailored to the specific design of this study.

4.4 Dataset 1 – Image Dataset

The first dataset used in this study is provided by Seungbae Kim. The dataset consists of 397 Instagram images sourced from Seungbae Kim's Instagram influencer dataset, which was originally collected for the study of "*Multimodal Post Attentive Profiling for Influencer Marketing*" (Kim et al., 2020). The images were collected from publicly available Instagram accounts and they represent consumer-generated content. Meaning, that the photographs are taken by real users in everyday setting rather than being brand-produced marketing materiel or runway images. This distinction is central to the analytical focus of this study, which is based on what consumers actually wear and how they present fashion social media rather than what brands choose to promote. The 397 images used for this study represent a sample from the family category, rather than the fashion category. This was done to test the fashion filter step of the pipeline, as not all images contain visible clothing or garments. Images without detectable fashion content were automatically rejected by the YOLOv8 fashion filter and excluded from further analysis.

The original dataset was collected over 92 days from October 1st 2018 to January 1st 2019, capturing posts with the hashtag #ad from 33,935 influencers across eight categories including beauty, family, fitness, fashion, food, interior, pet and travel.

The samples from the dataset were selected for three reasons. First, it provides organic generated images that aligns with the visual attribute extraction objectives of this study. Second, the data was collected systematically and documented transparently in peer-reviewed publication, making it credible and citable academic source. Third, the images are available for academic use, ensuring the analysis is reproducible and methodologically transparent.

The samples images serve as the primary input for the Visual Attribute Extraction Pipeline described in section 4.5. Each image was processed through a YOLOv8 model trained on DeepFashion2 weights, which then detected a total of 601 individual garments across the accepted images. The remaining images with no garments detected were rejected and excluded from further analysis.

This study acknowledges that images collected in 2018 may not fully reflect current fashion trends or contemporary consumer aesthetics. However, the purpose of this study is not to make claims on trends, but to demonstrate proof-of-concept by showing that the pipeline is capable of extracting meaningful visual fashion attributes from consumer-generated images. This limitation is discussed further in the Limitations subsection of Section 4.5.

4.4.1 Ground truth

A central challenge in evaluating the Visual Attribute Extraction Pipeline is that there does not exist an external ground truth for the dataset used in this study. The dataset only provides the images, but it did not provide any ground truth for garment category and sub-label. The outputs from the pipeline can therefore not be benchmarked against a pre-existing labelled corpus.

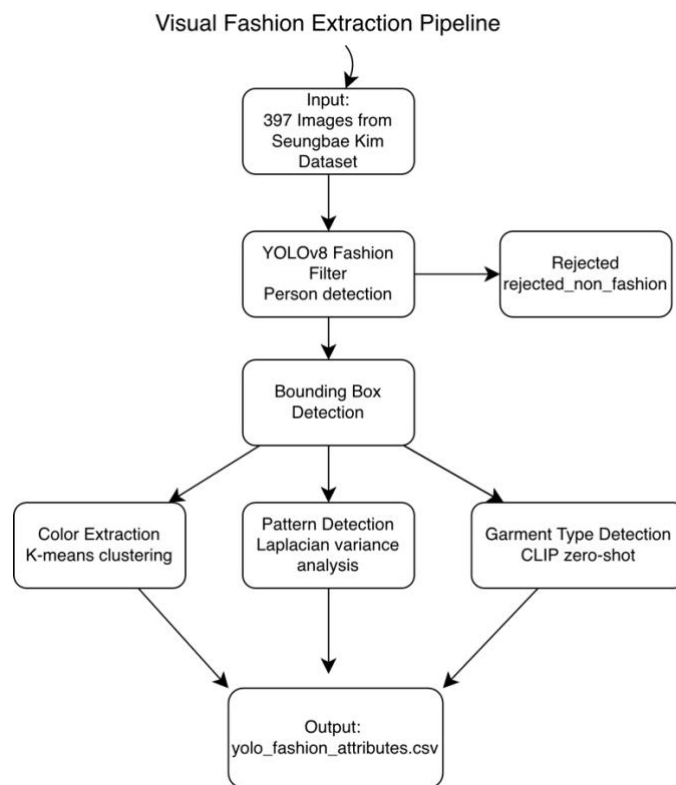
To address this challenge, this study constructed its own manual validation framework (`scr/validate.py`) that draws random images from the original dataset and presents them to a human evaluator for reviewing. The evaluator had to answer three questions for each image.

1. Is the bounding box accurate? (yes / partial / no). This evaluates the YOLOv8 detection stage.
2. What category does the garment belong to? (one of the 13 DeepFashion2 parent categories). This evaluates the broad classification.
3. What sub-category does the garment belong to? (one of the parent's CLIP sub-labels). This evaluates the fine-grained refinement.

The validation results will be presented in chapter 5: “Ground truth results”. The ground truth in this study for Pipeline 1 is measured as the agreement between the automated outputs and the human-assigned labels.

4.5 Section 1: Visual Attribute Extraction

This section presents the analytical pipeline for the Visual Attribute Extraction applied to the image dataset as described in section 4.4 (Kim et al, 2020). The purpose of the section is to introduce the different stages used to extract visual fashion attributes, specifically garment type, dominant colours, and pattern complexity from the sample images. The pipeline consists of four stages as illustrated in Figure 4.3.



“Figure 4.3: The Visual Fashion Extraction Pipeline. Own illustration created in app.diagrams.net”

Input

The first stage of the pipeline addresses a fundamental challenge in working with raw Instagram data. The dataset shows that not every image contains clothing. The raw Instagram data contains a mixture of content such as portraits, product flat-lays, scenery, memes, and everyday content where no garments are visible at all. This stage is directly motivated by RQ 1: *“What fashion attributes can be automatically extracted from images using YOLOv8, and what characterizes the visual content of consumer-generated fashion images?”*.

Before any attributes can be extracted from consumer-generated images using computer vision techniques, the pipeline must first identify which images contain fashion content and locate the individual garments within them. Similarly, H1 proposes that “*Visual fashion attributes extracted from images using automated computer vision techniques like YOLOv8 show distinct distributional patterns across garment types, colors, and patterns.*” Testing this hypothesis requires a clean and consistent set of garment detections as the foundation for all the extraction steps.

Stage 1 – Fashion Filter and Garment Detection

YOLOv8 and DeepFashion2 are introduced in Section 3.2; this stage applies them as following:

YOLOv8 runs on each of the 397 input images. For each of the images, the detector returns a set of candidate detections with a category label, a confidence score and a bounding box. A confidence threshold of 0.5, $conf_{thresh} = 0.5$, was applied to ensure that only detections with a confidence $conf_{score} \geq conf_{thresh}$ were kept. The images where no detections met this threshold were rejected and copied to a rejected_non_fashion folder. The images with at least one qualifying detection were accepted and passed to the next stage of the pipeline. The bounding box coordinates, garment category and confidence score for each accepted detection were recorded in a structured file called detections.csv. It is worth noting that a single image can produce multiple rows in the output since YOLOv8 is able to detect more than one garment in an image. An image that contains a person wearing a jacket and a pair of jeans. Each detection is recorded as a separate row in detection.csv, identified by the same image_id but with a unique garment_id.

The output for this stage is detections.csv:

```
image_id,garment_id,category,confidence,bbox_x,bbox_y,bbox_w,bbox_h
1917938963110614851.jpg,0,long_sleeved_shirt,0.9275985956192017,164.22056579589844,355.90625,608.4503326416016,440.5948486328125
1918481355538616589.jpg,0,trousers,0.8182454109191895,0.32814788818359375,571.6388549804688,631.6402969360352,340.08624267578125
1918481355538616589.jpg,1,long_sleeved_shirt,0.6709885001182556,195.34213256835938,438.9667663574219,204.69891357421875,201.52761840820312
```

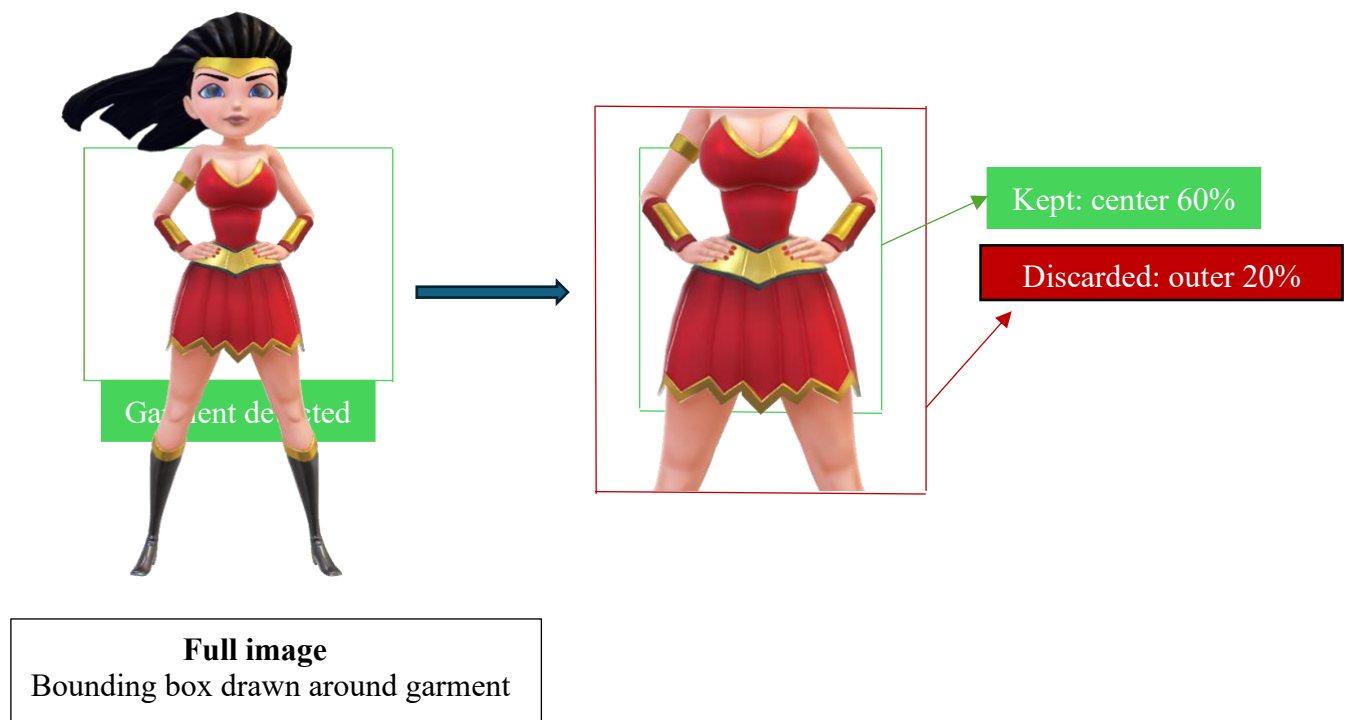
Table 4.1: Example output from Stage 1 – detections.csv. Each row represents one detected garment with its image identifier, garment index, category, confidence score and bounding box coordinates.

Stage 2 – Colour Extraction

K-means is introduced in Section 3.4; this stage applies it as following:

After the garments have been detected and bounding boxes recorded. The second stage extracts the dominant colour of each individual garment. Colour is one of the most vital tools in fashion for self-expression and visual impact. It sets the emotional tone of a garment, influences how others perceive the users, and it can show market trends. Identifying colour provides a signal on what consumers are wearing. And how they choose to present themselves (Labrecque, L. I., & Milne, 2012)

Rather than analyzing the colour of the entire input image, colour extraction is applied within the bounding box identified in Stage 1. This ensures that the analysis focuses on the garment instead of the surrounding area like background, hair, skin and sky etc. To further ensure this, a 60% crop is applied in the bounding box region, which discards 20% of the outer region. The aim is to reduce contamination from background pixels that can appear near the edges of a bounding box. Figure 4.7 illustrates this two-step process on how the bounding box detects followed by the 60% centre crop.



"Figure 4.7: Bounding box drawn around the detected garment (left). The 60% centre crop retains the inner region (green) and discards the outer edges (red), reducing background, skin and hair contamination before colour extraction. Own illustration."

The dominant colour in the detection box is matched to the nearest entry in a curated 20-colour fashion palette using Euclidean distance in CIELAB space. The palette was developed for this study to reflect commercially fashion colours and includes entries such as navy, burgundy, beige, denim, olive and khaki. This approach helps avoid generic colour names such as blue and red, which do not reflect fashion industry language. It furthermore changes RGB-values like (28, 38, 65) into "navy" and (160, 150, 100) into "khaki", this gives the garment a colourname.



"Figure 4.8: The 20-colour fashion palette used for colour extraction on the garments. Each garment's dominant colour is matched to the nearest palette entry using Euclidean distance in CIELAB space. Own illustration."

The output for this stage is color_attributes.csv:

```
image_id,garment_id,dominant_r,dominant_g,dominant_b,dominant_color_name,palette_rgb
1917938963110614851.jpg,0,59,65,57,brown,"[[59, 65, 57], [114, 105, 79], [219, 176, 125]]"
1918481355538616589.jpg,0,47,58,77,navy,"[[47, 58, 77], [180, 172, 143], [103, 108, 116]]"
1918481355538616589.jpg,1,39,54,77,navy,"[[39, 54, 77], [93, 117, 128], [60, 85, 101]]"
```

Table 4.2: Example output from Stage 2 – attributes.csv. Each row represents one detected garment with its dominant colour as RGB values, colour name from the fashion palette, and top-3 colour palette.

Stage 3 – Pattern Detection

Laplacian Variance is introduced in Section 3.5; this stage applies it as following:

Pattern detection is relevant for this study for two reasons. First, pattern complexity is a visually distinct attribute in fashion that cannot be captured by garment category or colour alone. Two garments can share the same garment category and dominant colour but look entirely different. To illustrate why pattern detection matters, consider two garments: a plain navy t-shirt and a striped navy t-shirt. In Stage 1 both garments would be classified as short sleeve top by YOLOv8. In Stage 2 the garments would return with navy as their dominant colour. This means that based on category and colour alone, the garments would appear as identical in the data. Yet to anyone who works in fashion, they would represent fundamentally different aesthetic choices. Pattern complexity helps capture a more accurate distinction on garments by assigning each garment a score that reflects how visually “busy” the fabric is.

Pattern complexity is furthermore relevant due to RQ1. Pattern, garment type and colour gives a more complete picture of the visual content, and they allow the results chapter to examine whether consumers in the dataset tend towards plain or patterned clothing and how this variance across garment types.

A plain and coloured garment produces very few edges and therefore a low Laplacian variance. In contrast, a patterned garment such as a floral dress or a striped shirt produces many edges and therefore a high variance. The variance of the Laplacian is used as a scalar measure of pattern complexity. The same 60% centre crop applied in Stage 2 is used here.

For this study a dataset-relative quantile threshold has been applied instead of applying fixed thresholds to classify garments. The 33rd and 66th percentiles of the variance scores across all 601 garments are computed, and each garment is assigned to one of three classes based on where its score falls. This means, that garments below the 33rd percentile are classified as plain, garments between the 33rd and 66th percentile as subtle and garments above the 66th percentile

as patterned. Using this approach ensures that the three classes are approximately equally distributed across the dataset and avoids arbitrary hardcoded values that would not generalize to a different dataset.

It is important to note that this method measures complexity only and it cannot distinguish between specific pattern types such as stripes, florals or checks. This means that a chunky knit sweater may also score high due to the texture of the weave rather than a printed pattern. These limitations are acknowledged and discussed further in the Limitations subsection of Section 4.5.

The output for this stage is `pattern_attributes.csv`:

```
image_id,garment_id,laplacian_variance,pattern_class
1917938963110614851.jpg,0,897.0200484315721,subtle
1918481355538616589.jpg,0,1193.2918592252936,patterned
1918481355538616589.jpg,1,486.81273891560335,subtle
```

Table 4.3: Example output from Stage 3 – `pattern_attributes.csv`. Each row represents one detected garment with its Laplacian variance score and pattern class (plain, subtle or patterned).

Stage 4 – Garment Type Refinement

CLIP is introduced in Section 3.3; this stage applies it as following:

The fourth stage of the Visual Attribute Extraction pipeline refines the broad garment categories introduced in Stage 1. DeepFashion2 provides the model with 13 garments categories, but they are too coarse for meaningful fashion analysis. Stage 1 is able to detect a garment as a short sleeve top but it says very little about what the person is actually wearing. A short sleeve top could be a graphic tee, a polo shirt, a blouse or even a crop top. Each of these tops represents a different consumer choice.

To address this limitation, this CLIP zero-shot classification is used to refine each broad category into a more specific sub-label. The specific CLIP model used is `openai/clip-vit-large-`

patch14, which has been accessed via Hugging Face transformers library. CLIP was selected for three reasons.

The first reason is it shows strong out-of-the-box performance on fashion categories without task-specific training. Second reason being, its zero-shot interface which allows the candidate label set to be edited in configuration rather than requiring retraining, this gives space for an exploratory pipeline where the taxonomy may evolve. Third, the model runs on CPU and is cached locally after the first download.

Parent-conditioned refinement

The CLIP refinement is conditioned on the parent category identified by YOLOv8 in Stage 1. This means, that a garment classified as long sleeve top is presented to CLIP with the candidates: sweater, hoodie, blazer, button-down short, turtleneck etc. and not labels from other categories. This is done to ensure that the refined label is consistent with the upstream detection.

CLIP returns a probability score for each candidate label after the prompt ensemble is averaged. The highest-scoring label is assigned as the refined garment type if its confidence meets or exceeds the parent's calibrated threshold. Garments falling below the threshold are marked as uncertain rather than being forced into a potentially incorrect category. This retains an explicit uncertainty class, important because zero-shot models can produce overconfident outputs even when their top-1 prediction is essentially a coin toss between visually similar candidates.

To illustrate how parent-conditioning and the calibrated threshold work in practice, consider two garments classified by YOLOv8 as `long_sleeved_shirt`. CLIP is asked to choose between the following candidates such as sweater, blazer, hoodie, button-down shirt, turtleneck, cardigan, and sweatshirt. If CLIP returns a top-1 probability of 0.34 for sweater, the garment is refined to a sweater. If the highest probability across all candidates is 0.18, the garment is marked as uncertain. The same top-1 probability of 0.34 under a parent using the default threshold of 0.40, for example “vest”, would instead be marked uncertain, because the parent's narrower vocabulary supports a more demanding standard..

The fashion taxonomy

The sub-labels were developed and organized into a hierarchical taxonomy covering all 13 DeepFashion2 parent categories, present in table 4.4. Labels were selected based on three criteria. First, visual distinctiveness in Instagram images. Second a prevalence in fashion language. Third a mutual exclusivity within each category. Sub-labels were selected for visual distinguishability in Instagram-quality images rather than for exhaustive coverage of fashion terminology. Labels that are visually indistinguishable from another label in the same parent or example, graphic tee versus t-shirt, or jumper dress versus pinafore were deliberately consolidated to avoid forcing CLIP to discriminate between visually equivalent candidates. This makes the labels more reliable.

DeepFashion2 Category	CLIP Sub-labels
Short sleeve top	T-shirt, polo shirt, blouse, crop top, tank top
Long sleeve top	Sweater, blazer, hoodie, button-down shirt, turtleneck, cardigan, sweatshirt
Short sleeve outerwear	Cropped jacket, light overshirt
Long sleeve outerwear	Trench coat, puffer jacket, leather jacket, wool coat, denim jacket, bomber jacket
Vest	Tailored vest, puffer vest, knit vest
Sling	Halter top, one-shoulder top, cami
Shorts	Denim shorts, tailored shorts, athletic shorts, bermuda shorts
Trousers	Jeans, tailored trousers, wide-leg pants, leggings, cargo pants, sweatpants
Skirt	Mini skirt, midi skirt, maxi skirt, pleated skirt, denim skirt
Short sleeve dress	T-shirt dress, mini dress, wrap dress
Long sleeve dress	Sweater dress, shirt dress, maxi dress
Vest dress	Slip dress, pinafore
Sling dress	Halter dress, one-shoulder dress, strappy dress

“Table 4.4: The fashion taxonomy mapping each of the 13 DeepFashion2 parent categories to its fine-grained sub-labels (53 sub-labels in total).”

The output for this stage is clip_refinement.csv:

```
image_id,garment_id,category_yolo,category_refined,refined_confidence,all_scores
1917938963110614851.jpg,0,long_sleeved_shirt,turtleneck,0.4619586765766144,{"t-shirt": 0.005775209981948137,
1918481355538616589.jpg,0,trousers,leggings,0.37172967195510864,{"jeans": 0.15667599439620972, "tailored tro
```

Table 4.5: Example output from Stage 3 – clip_refinement.csv. Each row represents one detected garment with its broad YOLOv8 category, refined CLIP sub-category, confidence score and full probability distribution across all candidate labels

Output

A final join-step combines the four different stages. Together they produce a CSV file, yolo_fashion_attributes.csv, keyed by image_id and garment_id. The file contains one row per detected garment with its full attribute vector. A left join is used and stems from Stage 1: Garment Category, detections.csv. This is treated as the source of truth for which garments exist. Every detected garment appears once in the final output, even if a downstream stage failed to produce an attribute for it. If the case appears, the missing value is stated explicitly.

The pipeline has been applied to 397 input images, the final output contains 601 rows with no missing values across any of the four attribute sources: Stage 1: Garment category, Stage 2: Colour Extraction, Stage 3: Pattern Detection and Stage 4: Refined Garment Type and confidence score.

The following table summarizes the columns included in yolo_fashion_attributes.csv

Column	Description
Image_id	Filename of the source image
Garment_id	Index of the garment within the image
Category	Broad garment category from DeepFashion2
Confidence	YOLOv8 detection confidence
Bbox_x, bbox_y, bbox_w, bbox_h	Bounding box coordinates
Dominant_r, dominant_g, dominant_b	Dominant colour as RGB values
Dominant_color_name	Colour name from palette
Palette_rgb	Top 3 colours as RGB values
Laplacian_variance	Raw pattern complexity score

Pattern_class	Plain, subtle or pattern
Category_refined	CLIP sub-category label or uncertain
Refined_confidence	CLIP confidence score

Table 4.6: Example output from the final join step– yolo_fashion_attributes.csv. Each row represents one detected garment with all attributes from the previous stages combined into a single structured dataset.

This file then serves as the primary dataset for the visual analysis and visualizations presented in the results chapter.

Limitations

Pipeline 1 has several limitations that should be acknowledged. One of the limitations in this dataset is the age and representativeness. The image dataset was collected in 2018 and consists of 397 images. The dataset is therefore relatively small and the findings should be interpreted as a proof-of-concept demonstration of the pipeline’s capabilities. It is important to state that the findings may not reflect current fashion trends, and a larger and more recent dataset would produce more robust and representative results.

Colour extraction is another limitation hence K-means clustering is sensitive to the number of clusters specified. K=3 was chosen as the practical balance between a simple and rich colour palette, but it may not capture the full colour complexity of patterned or multi-coloured garments. The 60% centre crop is able to reduce background noise, but cannot fully eliminate skin, hair or adjacent pixels from the analysis. The curated 20-colour palette, introduces a discretization step that may map visually distinct colours to the same palette entry. This means that two different shades of navy, both will be categorized as navy, even though they might look different.

Through the study, another limitation appeared. Pattern detection. The Laplacian variance method measures visual complexity as a proxy for pattern presence. The method can unfortunately not distinguish between specific pattern types like stripes or florals. It is important to note that the quantile-based thresholds are dataset-relative, this means that the

definitions of plain, subtle and patterned are adjust to this specific dataset. It can therefore not be directly compared across different dataset without adjustment.

Lastly, the CLIP zero-shot refinement. The classification here, is dependent on the quality of the candidate label vocabulary. Labels that are visually similar like sweater and cardigan, may produce low confidence scores and result in uncertain classifications. The CLIP refinement stage's uncertain rate of 4.16% reflects the combined effect of prompt ensembling and per-parent threshold calibration rather than an inherent limitation of the model.

The trade-off is one of methodological transparency by setting parent-specific thresholds, the refinement step accepts a small number of moderate-confidence predictions that a uniform high threshold would have abstained on. The alternative, a uniform 0.4 threshold across all parents would have produced an artificially inflated uncertain class concentrated in the parents with the largest vocabularies, which would have suggested a model limitation that is in fact a property of softmax over many candidates. The chosen calibration is the more honest of the two options, but does require trusting that thresholds at 0.25 are meaningful for parents with seven candidates above the uniform floor of 0.143.

4.6 Dataset 2 – Metadata Dataset

The second dataset is the Fashion Conversation Data on Instagram, obtained from Harvard Dataverse (<https://doi.org/10.7910/DVN/K7AW6F>). The dataset is composed of information about 24,752 posts by 24,072 unique users on Instagram, associated with 48 fashion brands across four brand categories. The near-one-to-one ratio between users and posts indicates that the dataset captures a single post per user in the large majority of cases, rather than repeated activity from a small number of accounts.

The categories in the dataset consist of High Street, Designer, Small Couture and Mega Couture. Unlike the image dataset, this dataset does not contain images. This dataset consists of metadata, and it is structured information about each post, this includes engagement metrics, pre-extracted visual content features derived from a Convolutional Neural Network model developed by the original authors. The behavioural and visual features such as Selfie and Marketing probabilities were obtained directly from the original dataset, which derived them

via image-analysis models developed by Ha et al. (2017). This study treats them as given input features rather than independently recomputing or validating them.

Table 4.7 presents the four brand categories and their associated brands:

Brand Category	Brands
Mega Couture	Burberry, Cartier, Chanel, Gucci, Hermés, Louis Vuitton, Prada, Tiffany
Small Couture	Brioni, Brunello Cucinelli, Cesare Attolini, Ermenegildo Zegna, Fabiana Filippi, Kiton, Loro Piana, Nancy Gonzalez
Designer	Acne Studio, Alexander McQueen, Alexander Wang, Coach, DVF, IRO, Isabel Marant, Kate Spade, Maison Margiela, Marc Jacobs, Michael Kors, Paul Smith, Rebecca Minkoff, Sandro, Stella McCartney, Theory, Vince, Vivienne Westwood
High Street	Abercrombie, American Apparel, American Eagle, Calvin Klein, Forever 21, Gap, Hollister, J.Crew, Mango, Topshop, Uniqlo, Urban Outfitters, Zara

The original dataset was collected over a month period in January, 2015. The posts were originally collected by searching Instagram for 48 brand-name hashtags such as #chanel or #topshop.

The metadata dataset was selected for three reasons. It provides a large-scale collection of consumer-generated Instagram posts, this supports the consumer behaviour classification objectives of this study. Second, the metadata includes engagement signals and pre-extracted visual features, enabling a multimodal analysis without additional image processing.

Lastly, the dataset is publicly available for academic use, ensuring that the analysis can be reproduced and methodologically transparent.

The dataset contains of 33 columns per post. Following table presents the 20 columns selected as relevant for this study. The classifier uses 19 engagement and visual features for the primary analysis. The column for hashtag will be used as a supplementary textual feature set. Table 4.8 represents the chosen columns.

Column	Role	Description
BrandCategory	Target	The brand category to predict
Likes, Comments	Feature: Engagement	Total number of likes and comments the post received
Followers	Feature: engagement (reach)	Number of users following the posting account. Retained as a feature to enable computation of reach_normalized engagement rates
MediaCount	Feature: engagement (reach)	Total number of posts the user has published.
Likes_per_follower	Feature: engagement (engineered)	Computed as Likes / (Followers + 1). Captures engagement intensity normalised by audience size.
Comments_per_follower	Feature: engagement (engineered)	Computed as comments / (Followers + 1). Captures comment intensity normalised by audience size.
Selfie,BodySnap,Marketing,ProductOnly,NonFashion,NumberOfFashionProduct,Face,Smile,Outdoor,Logo,BrandLogo,NumberOfPeople	Feature: Visual	Pre-computed probabilities and counts describing the visual content of each post.
Anger, Contempt, Disgust, Fear, Happiness, Neutral, Sadness, Surprise	Feature: Visual (emotion)	Pre-computed facial-emotion probability scores.
Hashtags	Feature: Textual (Supplementary)	List of hashtags in posts. Used in a supplementary analysis, as hashtags form a distinct textual modality outside the H2 comparison
Popularity_score	Descriptive (not a feature)	Computed as (Likes + comments) $\times$ 0.5. Reported alongside posts in popularity_score.csv but excluded from feature matrices, as it is a linear combination of

		Likes and comments which the model already receives as raw inputs.
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More information on the data is to be found in Appendix

All remaining columns such as user and post identifiers, image URLs and post links etc. were unrelated to the research objectives and excluded from the analysis.

Followers are retained as a feature alongside two engineered reach-normalised metrics, `likes_per_follower` and `comments_per_follower`, computed as $\text{Likes} / (\text{Followers} + 1)$ and $\text{comments} / (\text{Followers} + 1)$ respectively. The engineered rates respond directly to the follower-reach confound that complicates raw engagement metrics rather than excluding Followers. The pipeline encodes it as one feature among several and provides the classifier with both raw and reach-normalised forms of engagement. This allows the model to learn whether brand-tier signal lies in absolute reach, in engagement intensity, or in their interaction, rather than presupposing the answer through exclusion. MediaCount, the number of posts the user has published, is included on the same reasoning as a structural feature of the posting account rather than of the post itself.

BrandName was excluded from this analysis because it is too close to the target and could cause leakage. Caption was not used, as this study relies on hashtags.

The selected columns were chosen based on their relevance to RQ 2, H2 and H3.

RQ2 asks “*Can machine learning models classify Instagram content into fashion brand categories based on engagement and visual features, and what features drive this classification?*”. To answer this, the selected columns provide two principal feature types. Engagement features comprise the raw counts (Likes, comments, Followers, MediaCount) and the two engineered reach-normalised rates. Visual features comprise twelve pre-computed image-content probabilities (Selfie, BodySnap, Marketing, ProductOnly, NonFashion, Face, Logo, BrandLogo, Smile, Outdoor, NumberOfPeople, NumberOfFashionProduct) and eight facial-emotion probabilities.

H2 proposes that “A machine learning model combining visual and engagement features outperforms single modality models in classifying Instagram content into fashion brand categories”. To test this hypothesis, the feature set is partitioned into the engagement and visual groups described above, enabling each group to be evaluated independently and in combination.

H3 proposes that “*Consumer-generated fashion content displays distinct engagement patterns, and these patterns vary systematically across fashion brand categories*”. The visual features particularly the posting-behaviour probabilities (Selfie, BodySnap, Marketing, ProductOnly) and the emotion scores support a descriptive comparison of how consumers present fashion content across the four brand tiers, complementing the predictive analysis with a behavioural-pattern dimension.

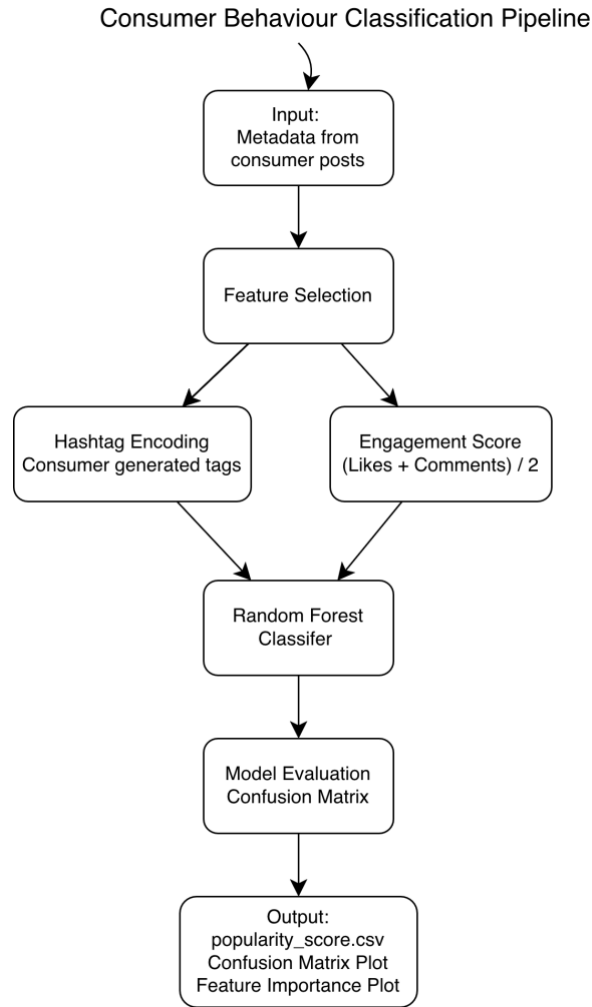
This study acknowledges that the dataset was collected in 2015 and it represents a cross-sectional snapshot of Instagram activity at a specific point in time. The dataset does not capture how engagement patterns change over time. This limitation is discussed further in the Limitations subsection of Section 4.7.

4.7 Section 2: Consumer Behaviour classification

This section presents the analytical pipeline for the Consumer Behaviour classification applied to the metadata dataset as described in section 4.6. The purpose of the section is to introduce the different stages used to examine how consumers use Instagram to engage with fashion brands, and whether a machine learning model can classify posts into fashion brand categories based on visual content features. The pipeline consists of four stages, illustrated in Figure 4.11.

Input

The second section of the analysis addresses a different but complementary challenge to Section 1. Unlike the image dataset, the metadata dataset does not require visual processing. The data here, has already been structured. All the data is however not equally relevant for the analysis. Using all posts without distinction would introduce noise into the classification model. The dataset is therefore filtered to retain only fashion relevant content, and the most informative columns were presented, justified and described in section 4.6. The following stages describe how the selected features are used to build and evaluate the pipeline



“Figure 4.11: The Consumer Behaviour Classification Pipeline. Own illustration created in app.diagrams.net”

Stage 1 – Data Preparation and Feature Selection

The first stage prepares the dataset for analysis. The 24,752 rows are loaded from the Excel file. Column names are stripped of leading and trailing whitespace, which renames the dataset's Comments column to comments. Rows with a null target are dropped; in practice, the dataset contains no such rows, but the check is retained for robustness.

The relevant columns identified in section 4.6 are selected as the analytical feature set. Following the definition of popularity in Jain and Singh (2025), a descriptive popularity score is computed for each post as:

$$Popularity_{score} = \frac{1}{2}(Likes + comments)$$

This score is recorded in `popularity_score.csv` alongside each post for reporting purposes, but is not used as a model feature. Because the score is a linear combination of Likes and comments both of which the classifier already receives as raw inputs including it would add no information beyond what is already present and would inflate the feature dimensionality without benefit.

The dataset exhibits substantial class imbalance, with Designer posts comprising 44.8% and Mega Couture only 7.5% of the sample. This motivates the use of balanced class weights during training and macro-averaged F1 as the primary evaluation metric, since overall accuracy would reward a model that simply ignored the minority class.

To prevent data leakage, all feature-engineering parameters including the hashtag vocabulary and the train-set statistics used for any normalisation are fitted on the training partition only and then applied to the held-out test set. A runtime assertion additionally verifies that excluded columns (BrandName, UserId, Link, ImgURL, Caption, CreationTime) never enter the feature matrix in any of the five ablation runs.

The output for this stage is `popularity_score.csv`:

```
BrandCategory,popularity_score,popularity_score_norm,Likes,comments,Followers,MediaCou
Designer,2.0,3.912669222943892e-05,4,0,1042,3376,0.000394491,0.00146354,0.00898937,0.9
High street,21.5,0.0004206119414664684,42,1,457,243,0.00247978,0.984923,0.0163281,0.00
Designer,15.5,0.00030323186477815165,30,1,2451,873,0.00480127,0.00399955,0.86896,0.036
```

Table 4.9: Example output from Pipeline 2, Stage 1 – popularity_score.csv. popularity_score.csv records each post with its computed raw and normalised popularity scores, its engagement and visual feature values, and a split column indicating its train or test partition.

Stage 2 – Hashtag Feature Extraction

The second stage extracts textual features from the hashtag column into numerical features that the classifier can process. Each post’s hashtag field is a single string of comma-separated tags, which is split into individual hashtags, lowercased, and stripped of surrounding whitespace and then leading “#” symbol. The posts with no hashtags are not discarded but retained and treated as an empty hashtag.

Stage 3 – Random Forest Classification

Random Forest Classification is introduced in Section 3.6; this stage applies it as following:

The third stage trains a Random Forest classifier to predict the brand category of each post from the selected features. Random Forest was selected for different reasons. The classifier handles mixed feature types and varying ranges well, which is useful for the feature set used in this study. It provides a built-in measure of feature importance, which allows the analysis to identify which features drive the classification, this is a requirement stated in RQ2. Lastly, Random Forest performs reliably on tabular data of this scale, and it is robust to irrelevant or correlated features. The dataset is relatively large and has a heterogeneous feature set.

The classifier is trained on three feature groups. Group A: engagement and reach comprises Likes, comments, Followers, MediaCount, and the two engineered reach-normalised rates (`likes_per_follower`, `comments_per_follower`). Group B: hashtags comprises the multi-hot encoding of the top-100 hashtags described in Stage 2. Group C: visual and behavioural comprises the twelve image-content features and the eight facial-emotion features supplied with the dataset.

Five ablation configurations are evaluated. Three test H2 directly: Group A alone, Group C alone, and their combination (the H2 test). Two further configurations are reported as methodological controls: Group B alone, and all three groups combined (A + B + C). The latter two are not part of the H2 test, because hashtags constitute a distinct textual modality and as the supplementary results will demonstrate the dataset's collection method via brand-name hashtags introduces brand-identity leakage into Group B. A runtime assertion guarantees that no hashtag-derived column enters the H2-combined feature matrix. Beyond the H2 comparison, the per-category feature importances and the differences in visual and engagement features across the four brands are examined to address H3.

Stage 4 - Model Evaluation & Behaviour Analysis

The purpose of the fourth stage, is to evaluate the trained classifier on the held-out test set, which contains posts the model has not seen during training. Given the substantial class imbalance, macro-averaged F1 is used as the primary evaluation metric. Macro-F1 computes the F1 score for each brand category independently and then averages them. In this way, each category contributes equally regardless of how many posts it contains. This prevents the evaluation from being dominated by the majority Designer class, and furthermore ensure that performance on the minority Mega Couture class is reflected in the headline result.

In addition to macro-F1, the evaluation reports per-class precision, recall and F1, as well as the confusion matrix. The confusion matrix shows how often each brand category is correctly classified and, where it is misclassified. This can be informative for interpreting consumer behaviour, as systematic confusion between two categories would suggest that consumers engage with those brand tiers in visually and behaviourally similar ways. Finally, the Random Forest's feature-importance scores are extracted to identify which features contribute most to the classification, which addresses the second part of RQ2.

The evaluation is structured as an ablation study. This design isolates the contribution of each of the feature groups. This is done by training the same model on different subsets of features and then comparing the results. This is done by holding the model, the train/test split, and all other settings constant. Meaning that only the input features changes between runs. This allows any difference in performance to be attributed to the feature group rather than to model or data split.

Output

Pipeline 2, namely Consumer Behaviour Classification produces several outputs. A processed dataset file records with the computed popularity score and selected features. A results file that summarizes the four classifier configurations and their evaluation metrics, enabling a direct comparison across the ablation runs. For each of the five configurations, a confusion matrix plot and a feature-importance plot are generated, alongside a per-class classification report. The trained model for the combined configuration is also saved. The outputs will form the basis for the visual and quantitative findings presented in the Results chapter.

Limitations

This pipeline contains several limitations that will be stated in the section. First, the visual and behavioural features were not computed by this study. The features were obtained directly from the original dataset, where they were derived by image-analysis models by Ha et al. (2017). It is important to mention that their accuracy is taken as given and not independently validated. This means that any error in the original feature extraction spreads into this analysis. No content-based filtering is applied to the dataset; the NonFashion probability is retained as a visual feature and used by the classifier rather than as an exclusion criterion. This means the classifier is presented with the full distribution of posts in the dataset, including those the original authors' image-analysis models flagged as low-confidence fashion content.

Engagement-based features must also be interpreted with caution, as the number of likes and comments a post receives is influenced by different factors such as follower base rather than by post content alone. This is partly why follower count is excluded as a feature, although the confound cannot be entirely eliminated.

The dataset represents a cross-sectional snapchat and the findings should not be interpreted as evidence of temporal trends.

Another limitation is the hashtag group that almost encodes brand identity directly. This is reflected in the strong performance of the hashtag-only ablation run and is the reason hashtags are treated as a supplementary and excluded from the primary H2 evaluation.

Lastly, categories are assigned by brand, and not by what's actually in the post. A post can therefore be labelled "Mega Couture", because of the hashtag #Gucci. This is regardless of the content. This introduces a fundamental yet interesting take on what the classification task is actually measuring. A user posting an image with the #Gucci may be wearing Gucci garment, tagging it ironically or simply use #Gucci as a visibility hashtag to reach a broader audience. As a result, the classifier learns to predict which brand a post is associated with rather than how consumers exactly express or engage with fashion. This limits the extent to which the results can be interpreted as direct evidence of consumer behaviour, and it is important to note when addressing H3.

4.8: Environmental Design and Reproducibility

All analysis and modelling pipelines were implemented in Python 3.13. Dependency management and command execution were handled with uv, which was used to create the project environment, lock package versions, and run all pipeline scripts. The repository contains the source code, configuration files, dependency lockfile, and pipeline entry points required to reproduce the conducted experiments. The project is available at:

<https://github.com/jellalab/FashionTrendPrediction>

5 Results

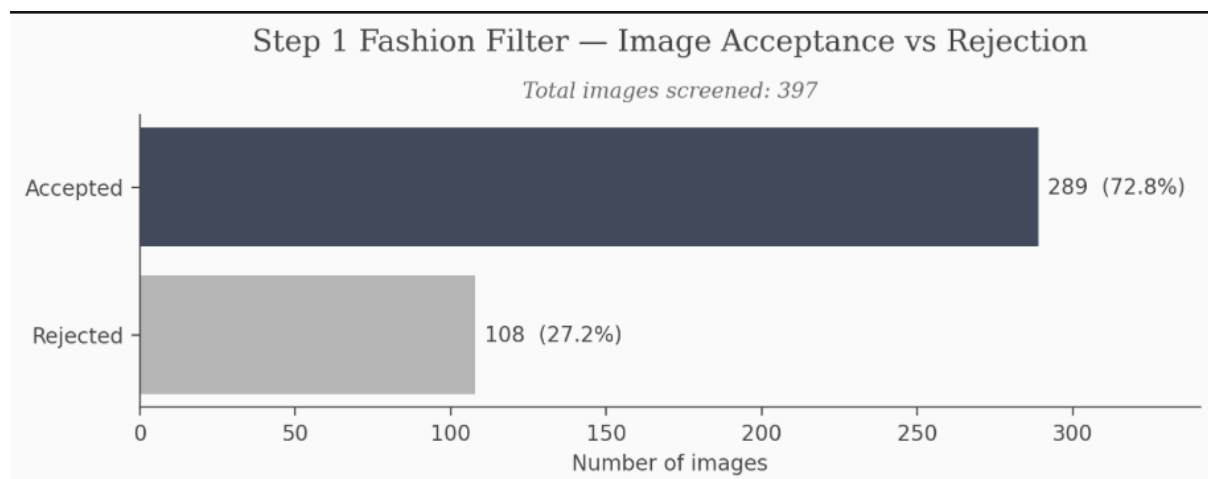
What the Algorithm Saw

Chapter 4 formulated the methodology applied in this study to address the problems presented by the research questions. This chapter presents the performance of the two analytical frameworks, considering the research objectives. Section 5.1 presents the results of the Visual Attribute Extraction Pipeline applied to the image dataset. Section 5.2 presents the results of the Consumer Behaviour Classification Pipeline applied to the Instagram metadata dataset.

5.1 Visual Attribute Extraction Results

5.1.1 Fashion filter and garment detection

The fashion filter and garment-detection stage operationalises the first half of RQ1: *what fashion attributes can be automatically extracted from images using YOLOv8?* The output is a structured table of garment-level detections with parent categories, confidence scores, and bounding-box coordinates, which downstream stages build upon. As seen in figure 5.1



“Figure 5.1: Fashion filter – Image acceptance vs Rejection”

By applying 397 input images, the fashion filter accepts 289 images (72.8%), and rejects 108 (27.2%), input images. 601 individual garments were detected across the accepted images, equal to a mean of 2.08 garments per accepted image.

The most frequently detected categories are long sleeve top ($n = 221$), trousers ($n = 179$), and short sleeve top ($n = 105$), with a longer tail across dresses, outerwear, vests, skirts, and sling garments (each ≤ 20 detections). Together, these three dominant categories account for 84.0% of all detections.

The distributional pattern directly supports the first half of H1: garment categories show clearly distinct frequencies rather than a uniform spread. Three categories dominate, and the tail is long and sparse. The dominance of upper-body garments is consistent with the visual conventions of consumer-generated Instagram content, where torso-up framing is the most common compositional choice such as selfies, mirror shots, and head-and-shoulders portraits all preferentially capture upper-body clothing while cropping or partially obscuring lower-body garments. The fact that trousers nonetheless rank second suggests that full-body and waist-down compositions also occur with meaningful frequency, but at roughly half the rate of long-sleeve tops.

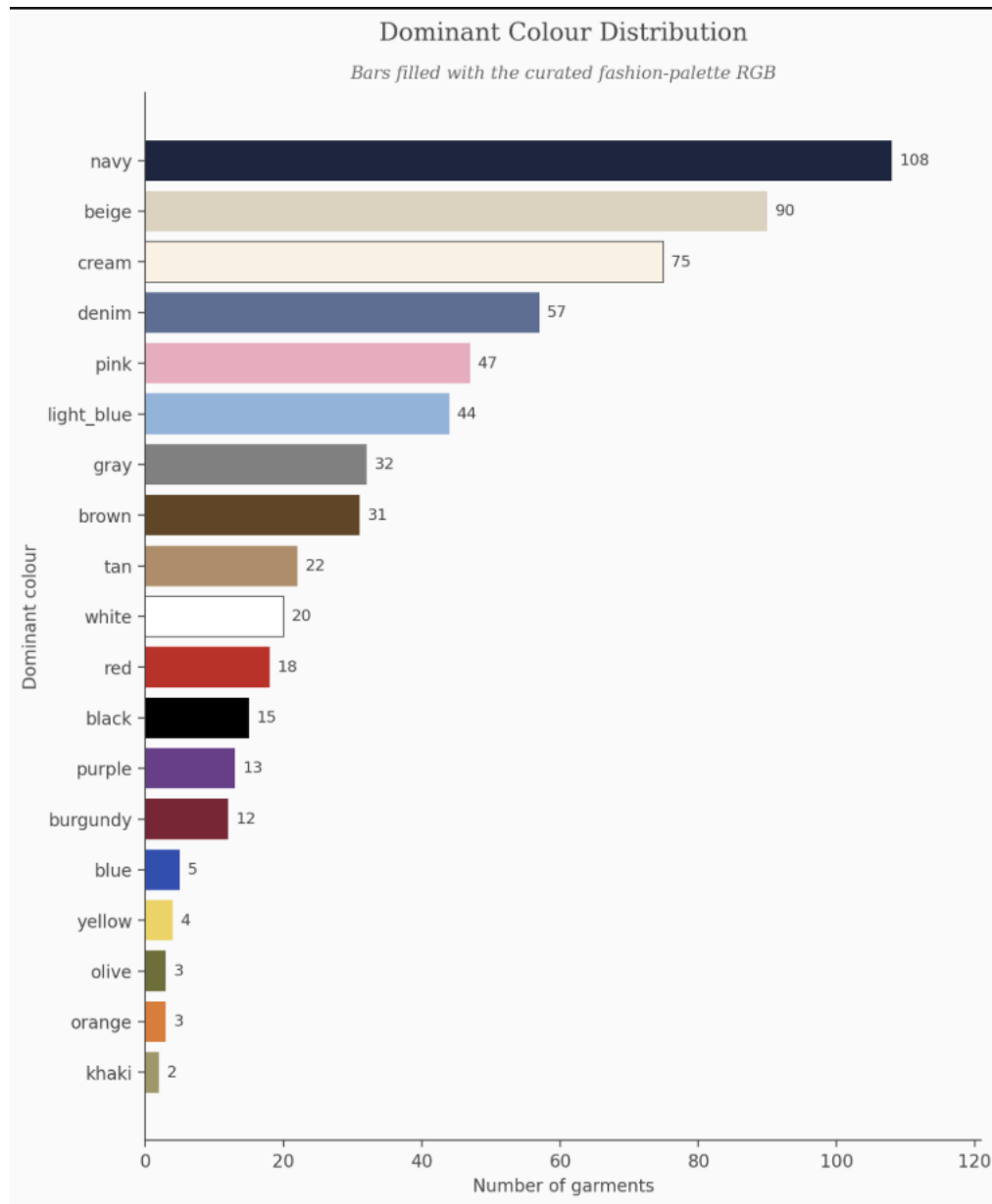
The scarcity of dresses, skirts, and outerwear (each ≤ 20 detections) more likely reflects the dataset's composition drawn from the family category of the Kim et al. (2020) dataset rather than the fashion category than a true absence of these garments from consumer Instagram content more broadly. The methodological consequence is that downstream attribute analyses are statistically well-supported only for the three dominant parent categories; findings concerning the rare-parent tail must be interpreted as suggestive rather than conclusive.

The 27.2% rejection rate is itself a methodologically meaningful figure. It indicates that more than a quarter of the input images contained no detectable garments at the chosen confidence threshold of 0.5 a finding consistent with the observation in section 4.4 that the sample was drawn from the family category rather than the fashion category, and that the YOLOv8-DeepFashion2 model returned no qualifying detections for posts dominated by faces, objects, scenery, or other non-garment content.

5.1.2 Dominant colour distribution

The dominant-colour distribution describes the chromatic content of the detected garments. For each garment, the dominant colour is extracted via K-means clustering in CIELAB colour

space within the 60% centre crop of the bounding box (section 4.5 Stage 2), then mapped to the nearest entry in a curated fashion palette by perceptual distance.



“Figure 5.2: Colour clustering – Dominant Colour Distribution”

Colour extraction was applied to the 601 detected garments. The most frequently detected colour is navy at 17.97% (108 of 601 garments), followed by beige at 14.98% (90 garments) and cream at 12.48% (75 garments) (Figure 5.2). When the three blue-family palette entries are combined, navy, denim, and light_blue, they account for 34.78% of all garments (209 of 601). Together with the warm-beige entries (beige, cream, tan, khaki), neutral and earth tones constitute the majority of the dataset.

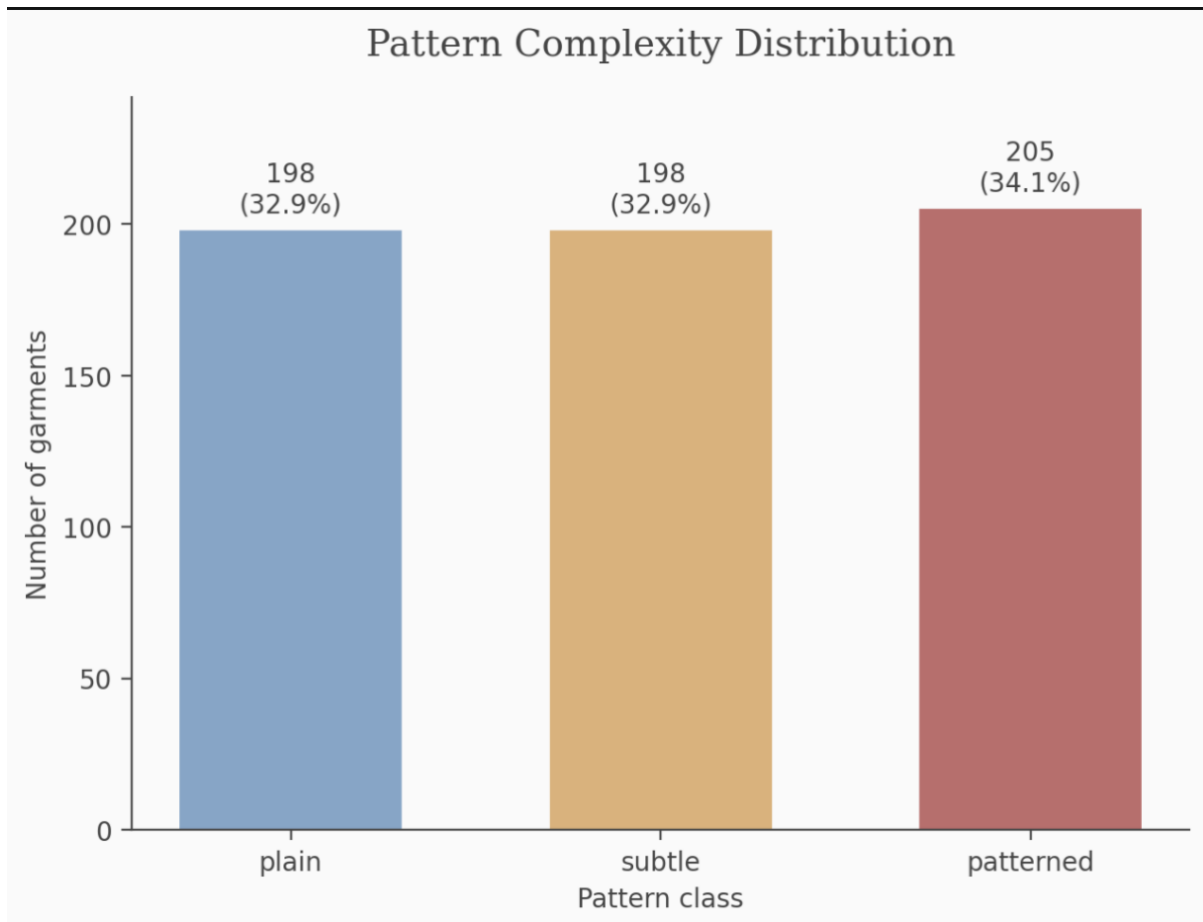
Saturated colours are notably rare. Yellow, olive, orange, and khaki each appear on four or fewer garments. This is a meaningful finding for RQ1: the visual content of consumer-generated fashion images is dominated by muted tones rather than the high-saturation colours that characterise runway or marketing imagery. Even within the broader neutral family, the data show that consumers gravitate toward the cool-blue and warm-beige axes of the palette, with mid-saturation colours such as olive and khaki both nominally fashionable appearing far less frequently than their warmer or cooler neighbours.

This pattern supports H1's claim of distinct distributional patterns across colours: colours are not uniformly distributed but form a clear hierarchy in which neutrals dominate and saturated tones occupy the long tail. The finding is also consistent with the dataset's contextual provenance everyday consumer photographs taken in domestic and outdoor settings rather than commercially styled fashion shoots — where wardrobe choices tend toward functional neutrals.

The dominant-colour extraction is performed in CIELAB colour space after a 60% centre crop of each bounding box, which means the extracted colour reflects the perceptually salient portion of the garment rather than its edge regions. The strength of the neutral concentration therefore cannot be attributed to background or skin contamination of the crop, but reflects a genuine property of the garments themselves.

5.1.3 Pattern complexity distribution

The pattern-complexity stage classifies each garment as plain, subtle, or patterned based on Laplacian-variance scoring of its centre-cropped greyscale region (section 4.5 Stage 3). Quantile thresholds at the 33rd and 66th percentiles of the variance distribution were used in place of fixed absolute thresholds, to ensure the classification generalises without dataset-specific tuning.



“Figure 5.3: Pattern – Pattern Complexity Distribution”

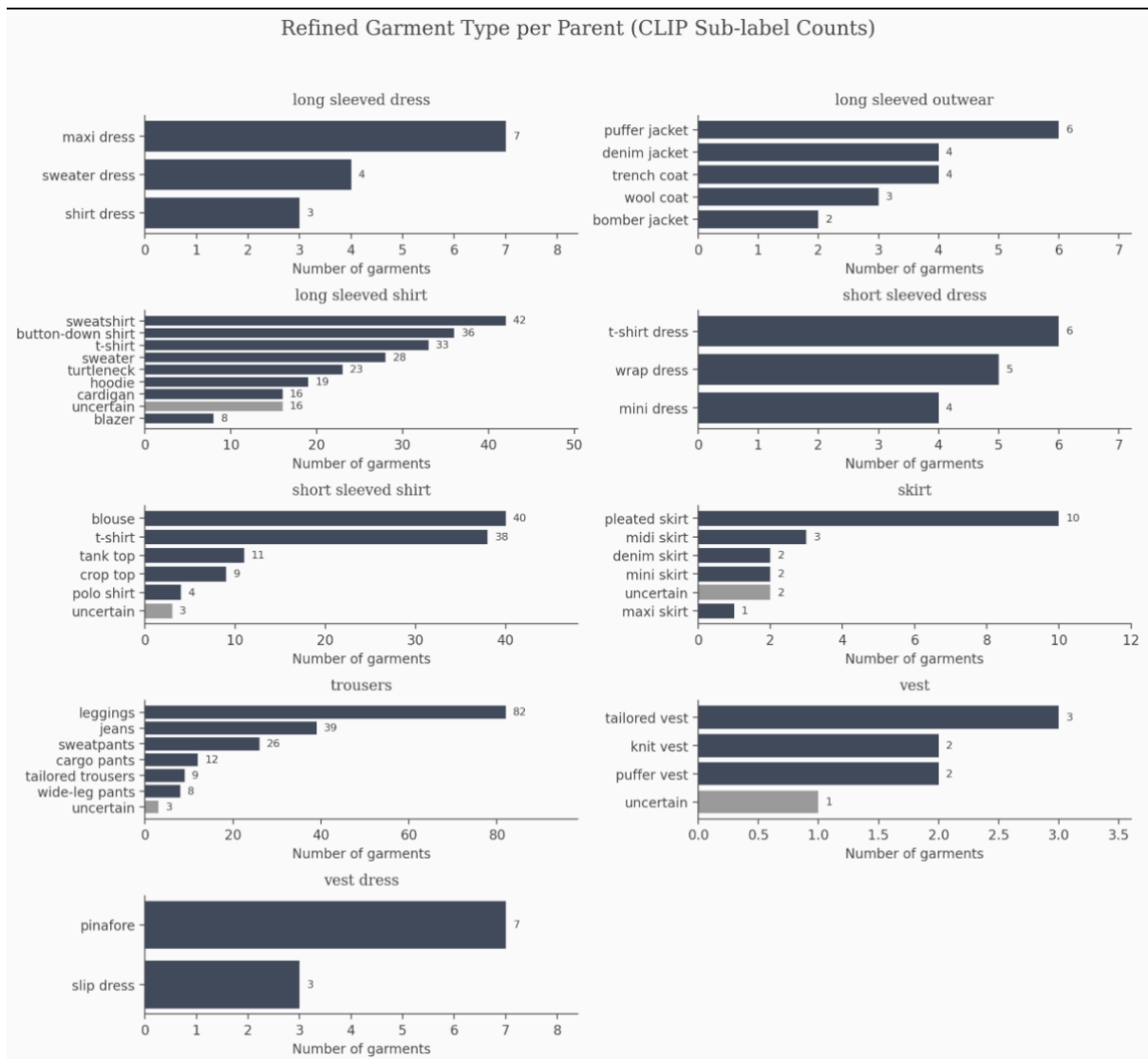
Pattern detection was applied to the 601 detected garments, producing an approximately even distribution across the three classes: 32.9% plain (198 garments), 32.9% subtle (198 garments), and 34.1% patterned (205 garments) (Figure 5.3). This near-perfect balance is, importantly, a direct consequence of the methodology rather than a property of the data: bucketing variance scores by the 33rd and 66th percentiles by construction produces approximately equal class sizes. Any dataset of any composition would yield a roughly 33/33/33 split under this scheme.

The methodological choice to use quantile thresholds was made in section 4.5 to avoid arbitrary hardcoded values that would not generalise across datasets. The trade-off is that the resulting class distribution does not, on its own, describe the data: the informative analysis of pattern complexity lies not in the marginal distribution of pattern classes, but in their relative variance scores and in their interaction with other attributes.

For H1, the pattern dimension therefore provides weaker evidence than the garment category or colour dimensions: while the underlying continuous Laplacian-variance scores do vary systematically, the categorical pattern_class output does not, by methodological design, exhibit a "distinctive" distribution in the same sense as categories or colours. The pattern dimension's analytical value lies in its capacity to differentiate garments that share the same garment category and dominant colour for example, a plain navy long sleeve top from a patterned navy long sleeve top rather than in its marginal distribution.

A further methodological caveat applies: Laplacian variance measures visual complexity, not pattern type. A high-variance garment may be a floral dress, a striped sweater, or a chunky-knit jumper whose texture rather than print produces the edge density. The constraint is acknowledged in section 4.5 and limits the interpretive weight that can be placed on the pattern-class distribution alone.

5.1.4 CLIP refinement



“Figure 5.4: Refined Garment Type per Parent (CLIP Sub-label Counts)”

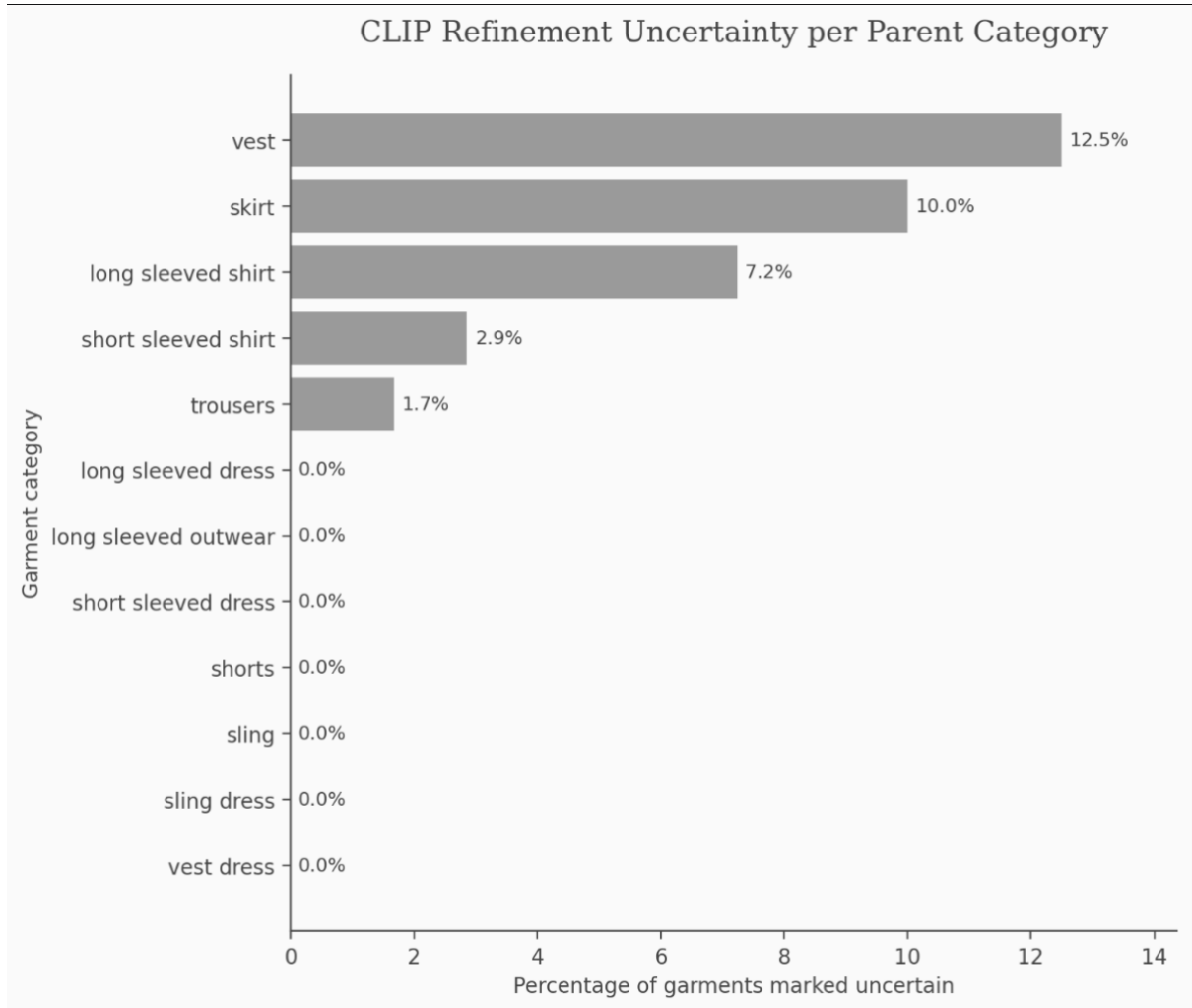
Figure 5.4 is a distribution of CLIP-refined sub-labels within each DeepFashion2 parent category that contained more than five detections. Each subplot shows the count of garments assigned to each sub-label, sorted by frequency. Uncertain assignments are garments whose top-1 CLIP probability fell below the parent's calibrated threshold are shown in grey to distinguish them from confidently refined sub-labels.

Figure 5.4 shows how the CLIP refinement distributes garments across sub-labels within each parent category, making visible that the refinement step does meaningful analytical work rather than collapsing all garments in a category onto a single dominant sub-label, the distributions reveal varied assignments that the broad DeepFashion2 categories alone could not surface.

Within the three dominant parent categories, clear modal sub-labels emerge. The figure shows how three garment types accounts for 84% of everything detected which are long sleeve shirt, trousers and short sleeve shirt

In the long sleeve shirt category ($n = 221$), sweatshirts are the most frequent sub-label at 42 garments (19%), followed by the remaining six candidates in the long-sleeve vocabulary. Within trousers ($n = 179$), leggings are by far the most frequent sub-label at 82 garments (46%), accounting for almost half of all detected trousers. Within short sleeve shirt ($n = 105$), blouses dominate at 40 garments (38%). The relative size of the uncertain bar within each parent gives an immediate read on where the model is least confident: uncertainty is concentrated in only three parents (vest, skirt, long sleeve top) and is absent from the remaining ten, as illustrated in figure 5.5.

Taken together, the three modal sub-labels, where sweatshirts dominate long sleeve shirt, leggings dominate trousers, and blouses dominate short sleeve shirt sketch a consistent aesthetic register across the dataset's most-detected categories: casual, comfort-oriented dressing rather than tailored or formalwear. This pattern is consistent with the contextual provenance of the dataset, drawn from the *family* category of the Kim et al. (2020) Instagram collection rather than the *fashion* category. The dominance of sweatshirts and leggings in particular suggests a substantial presence of athleisure or domestic-comfort content, while the prominence of blouses in the short-sleeve category indicates that consumers' "dressed-up" choices in this dataset trend toward soft tailoring rather than formal shirting.



“Figure 5.5: CLIP Refinement Uncertainty per Parent Category”

The CLIP zero-shot refinement stage assigned a fine-grained sub-label to 576 of 601 garments (95.84%) and marked 25 garments (4.16%) as uncertain. The uncertain rate is markedly lower than would be expected from a naive single-template, single-threshold application of CLIP to a hierarchical fashion taxonomy, and reflects two deliberate methodological choices described in section 3.3 and section 4.5: prompt ensembling and per-parent threshold calibration.

The uncertain rate is not uniformly distributed across the 13 parent categories. The highest rate occurs in vest at 12.5%, followed by skirt at 10.0%, with long_sleeve_shirt at 7.2% a category whose seven-candidate vocabulary would, under a uniform 0.4 threshold, produce a much higher abstention rate by construction. The seven remaining parent categories show 0% uncertain. The figures for the rare parents must be read with caution: vest contains 8 detections in total, so a 12.5% rate corresponds to a single garment.

This pattern has two implications for RQ1. First, the refinement step delivers value disproportionately in the parents with the largest detection counts (long-sleeve top, trousers, short-sleeve top), which are also the categories where the broad DeepFashion2 label leaves the most analytical ambiguity. Second, the small remaining uncertain class concentrated in just three categories represents genuine cases where a single Instagram photograph does not contain enough visual evidence to distinguish between visually similar sub-categories, even with calibrated thresholds and ensembled prompts. This is the correct behaviour for a methodologically honest refiner rather than abstaining widely under the safest possible threshold, the system abstains narrowly at the points where the visual evidence actually runs out.

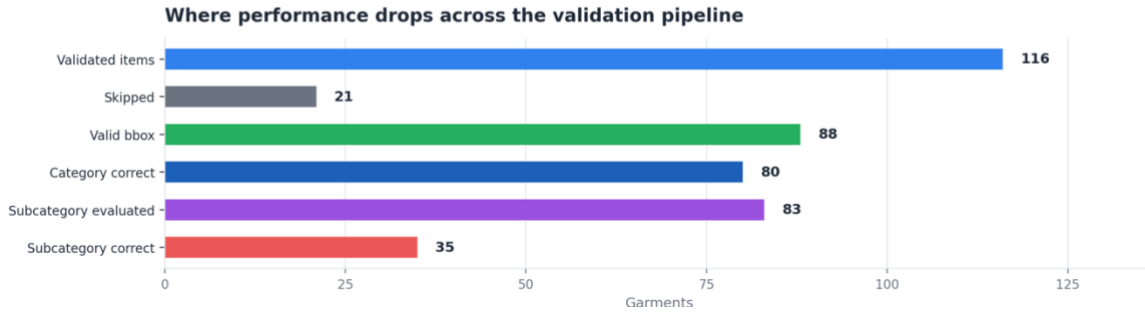
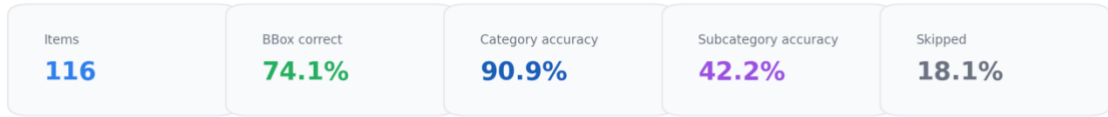
The mean refined-confidence score across all garments is 0.557. This indicates that, while the model produces a refined label for the large majority of garments, the typical confidence sits at a moderate rather than high level consistent with the inherent visual similarity between adjacent sub-categories in fashion (sweater vs. sweatshirt, trench vs. wool coat) and with CLIP's pretraining on a broad rather than fashion-specific corpus.

In support of H1, the refined sub-label distribution adds meaningful analytical depth within each parent category beyond what is available from DeepFashion2 alone. Combined with the consistent application of per-parent thresholds and the four-template prompt ensemble, the refinement stage produces a structured fine-grained classification of consumer-generated fashion content that satisfies the second half of RQ1: what characterises the visual content of consumer-generated fashion images?

Ground Truth

Validation summary

A compact view of localization, category accuracy, subcategory accuracy, and uncertainty.



“Figure 5.6: Validation summary”

To test the pipeline’s accuracy a validation application was constructed as described in section 4.5.5. An external evaluator reviewed 116 randomly selected garments from the image dataset and had to assign them to three dimensions from the pipeline’s output.

1. Bounding box accuracy
2. Parent category agreement
3. Sub-category agreement

137 images were reviewed in the validation framework and 21 images (18.1%) were skipped by the evaluator which could be because the garments were out of frame to a degree that prevented a confident judgment, ambiguous or occluded. Figure 5.6 presents the outcomes across the different stages.

The validation results show a clear pattern. The pipeline accuracy declines as the precision of the classification increases. Bounding-box localisation was judged accurate in 74.1% of cases (88 of 116). Parent-category assignment, the broad DeepFashion2 classification, agreed with human judgment in 90.9% of cases where a bounding box was valid (80 of 88). Sub-category refinement, the finest-grained output produced by the CLIP zero-shot stage, agreed with human judgment in only 42.2% of cases where evaluators rendered a sub-category judgment (35 of 83).

The high parent-category accuracy (90.9%) provides direct empirical support for the YOLOv8-DeepFashion2 detection stage and for the methodological choice to treat its outputs as reliable parent labels for downstream stages. The model *can* be trusted to identify whether a garment is, for example, a trouser rather than a skirt, or a long-sleeve top rather than outerwear. The core categorical distinctions that anchor the rest of the pipeline.

The substantially lower sub-category accuracy (42.2%) is a more challenging finding. It indicates that within a correctly-classified parent category, the CLIP refinement produces a sub-label that human evaluators judge correct less than half the time. Several factors might contribute to this gap. First, sub-category distinctions are inherently more difficult than parent distinctions for instance telling a sweatshirt from a sweater, or leggings from sweatpants, requires fine visual cues that may not be present in a single Instagram photograph at the resolution available. Second, the same ambiguity that affects the pipeline also affects the evaluators, and the way the validation measures agreement rather than correctness means that any garment for which the evaluator and the pipeline reasonably reach different conclusions registers as pipeline error, even where neither answer is uniquely correct.

The CLIP model used in this study was not fine-tuned on fashion imagery; it is a general-purpose vision-language model whose familiarity with fine-grained fashion taxonomy is bounded by what appears in its pretraining corpus.

The 74.1% bounding-box accuracy sits between the two classification figures and reflects the YOLOv8 detector's tendency to produce loosely-fitting boxes in cluttered scenes, occluded garments, or images where multiple people overlap. Although a loose bounding box does not necessarily invalidate downstream colour and pattern extraction (the 60% centre crop mitigates moderate over-extension), it does represent a localisation limitation that downstream stages cannot fully recover from.

Taken together, the validation results support a differentiated reading of the pipeline's capabilities. The pipeline is accurate at the level of what kind of garment is in the image (parent category) and reasonably accurate at where the garment is (bounding box), but is markedly less accurate at what specific sub-type of garment it represents. This pattern aligns with the design hierarchy of the pipeline: YOLOv8 plus DeepFashion2 provide a robust detection-and-classification foundation, while the CLIP refinement adds optional finer granularity at the cost of substantially reduced accuracy. Downstream analyses in this study and discussions of findings should be read with this accuracy hierarchy in mind: claims at the parent-category level are strongly supported, while claims at the sub-category level are exploratory.

5.1.5 Summary of findings

Across the four attribute dimensions, H1 receives differentiated support. The garment-category and dominant-colour distributions provide strong evidence of distinct distributional patterns. Both show pronounced concentration in a small number of dominant entries with a long sparse tail, consistent with H1's prediction. The CLIP refinement stage adds analytical depth within each parent category and supports H1 through the richer sub-label distributions, while transparently quantifying its own uncertainty through the per-parent threshold mechanism. The pattern-class dimension provides the weakest evidence for H1, but for a methodological rather than substantive reason. The quantile-bucket design produces a near-uniform marginal distribution by construction. The interpretive value of pattern complexity therefore lies in its interaction with other attributes, addressed in the Discussion chapter, rather than in its marginal distribution alone. Taken together, the visual-attribute pipeline successfully extracts a structured, multidimensional characterisation of consumer-generated fashion content from raw Instagram images, addressing RQ1 directly and providing partial-to-strong support for H1 across the four dimensions examined.

5.2 Consumer Behaviour Classification Results

5.2.1 Class distribution

The support column from the classification reports gives the class distribution for the test set. It looks as following:

Designer	2218
Small Couture	1207
High Street	1155
Mega couture	371
Total	4951

Tabel 5.2.1 presents the class distribution

The metadata dataset contains 24,752 posts across four brand categories. Following an 80/20 train-test split, the test partition contains 4,951 posts. The class distribution is substantially imbalanced across the four categories. Designer posts constitute the largest share at 2,218 posts (44.8%), followed by Small Couture at 1,207 posts (24.4%), High Street at 1,155 posts (23.3%), and Mega Couture at 371 posts (7.5%).

This imbalance has direct methodological consequences. A classifier that simply predicted Designer for every post would achieve an accuracy of 44.8% without learning any meaningful pattern. Overall accuracy is therefore an unreliable evaluation metric for this dataset. This motivates two design choices described in section 4.7: the use of balanced class weights during training to prevent the model from over-predicting the majority class, and macro-averaged F1 as the primary evaluation metric, which weights each brand category equally regardless of how many posts it contains.

5.2.2 Ablation results

The classification pipeline was evaluated under the five ablation configurations described in section 4.7, three of which constitute the formal test of H2 and two of which serve as a methodological control surfacing the dataset's brand-hashtag leakage.

Run	Features	Accuracy	Macro-F1
Group A: Engagement	6	0.633	0.619
Group B: Hashtags	101	0.959	0.938
Group C: Visual/behavioural	20	0.545	0.492
Combined engagement + visual (H2 test)	26	0.630	0.591
Combined all features (with hashtags)	127	0.957	0.949

Table 5.2.2 presents the accuracy and macro-F1 across five ablation configurations described in section 4.7.

The three runs that test H2 are engagement only, visual/behavioral only and their combination. The three runs fall within a macro-F1 range of 0.49 to 0.62.

Engagement features alone achieve the highest macro-F1 at 0.619. The visual/behavioural feature group alone achieves 0.492. The combination of both achieves 0.591. The combined model does not outperform the best single-modality model. H2, which predicts that combining visual and engagement features outperforms single-modality models, is therefore not supported by the data. Adding visual/behavioural features to engagement features slightly degrades performance rather than improving it. This is a substantive finding: in this dataset, the visual content of a post and the engagement it receives do not combine additively, and engagement carries more discriminative information about brand tier than the visual content does.

The two runs involving hashtag features are the hashtag-only run and the combined-all-features run. They achieve substantially higher macro-F1 scores (0.938 and 0.949

respectively), more than 0.3 points above the strongest non-hashtag run. This gap is not a model success but a methodological finding about the dataset. The Kim et al. (2020) data were collected via Instagram searches for brand-name hashtags such as #chanel, #zara, and #topshop, meaning that the hashtag field of each post almost invariably contains a tag that encodes the brand identity directly. When hashtags are used as features, the classifier is effectively reading the brand off the post rather than learning a content-based or behavioural association. The five-run ablation surfaces this leakage empirically and justifies the methodological decision to exclude hashtag features from the H2 test (section 4.7 Stage 3).

The same logic applies, by extension, to any column that encodes brand identity rather than the consumer behaviour around it. BrandName was excluded a priori for the same reason; hashtags were retained only for the supplementary leakage-demonstration runs; and a runtime assertion hard-blocks any hashtag-derived column from entering the H2 combined feature matrix. The clean H2 comparison therefore relies only on features that describe what consumers post and how others engage, not which brand they tag.

5.2.3 Confusion matrix interpretation

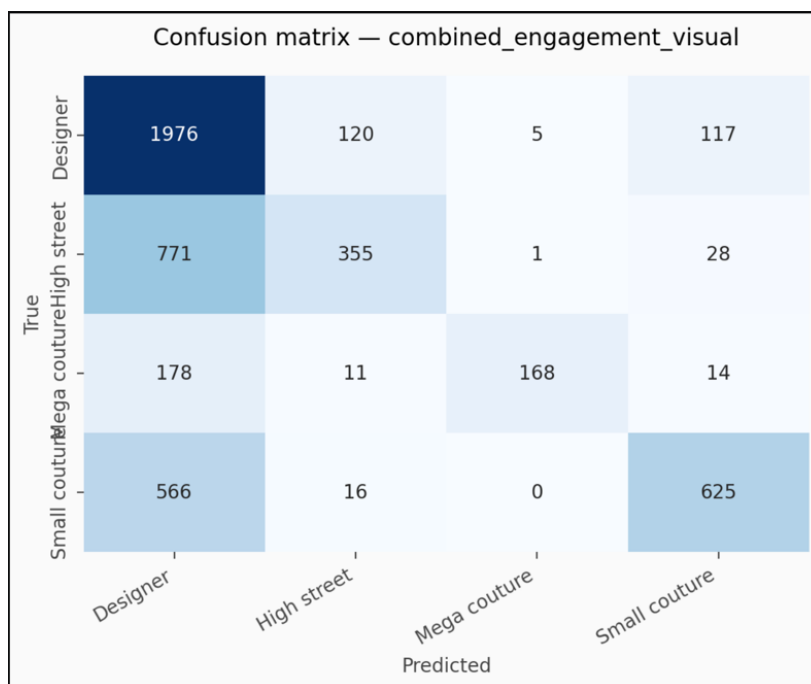


Figure 5.2.1 presents the confusion matrix for combined engagement and visual

Figure 5.2.1 presents the confusion matrix for the H2 test configuration (engagement + visual/behavioural features). The matrix reveals a dominant error pattern that explains the macro-F1 results.

Designer is classified accurately, with 1,976 out of 2,218 Designer post (89.1%) being correctly identified. But, the three other brands show substantial misclassifications and the errors are not distributed randomly.

Across every other class, the most common misclassification is into Designer. High Street is the most severely affected: of 1,155 High Street posts, only 355 (30.7%) are correctly classified, while 771 (66.8%) are predicted as Designer. Small Couture follows a similar pattern, with 566 of 1,207 posts (46.9%) misclassified as Designer; correct Small Couture classifications stand at 625 (51.8%). Mega Couture, despite being the smallest class, is correctly classified in 168 of 371 cases (45.3%), with 178 (48.0%) misclassified as Designer.

The error pattern is analytically informative. First, it tells us that the model is heavily biased toward predicting Designer, majority class at 44.79% of the dataset, despite the use of balanced class weights during training.

This suggests that the visual and engagement signal available to the model is genuinely insufficient to separate brand tier cleanly: when in doubt, the model defaults to the largest class. Second, the errors are asymmetric in a meaningful way. Mega Couture is misclassified almost exclusively as Designer (178 of 203 errors) it is essentially never confused with Small Couture (14 cases) or High Street (just 1 case). This asymmetry directly informs H3: Mega Couture posts are clearly distinct from the lower brand tiers along the dimensions the model can see, but are visually and behaviourally adjacent to Designer posts. The luxury–mass distinction is recoverable from the data; the gradations between Mega Couture, Designer, and Small Couture luxury are not.

Designer's high accuracy is therefore partly a property of model bias (a class predicted often will be predicted correctly often) and partly a property of genuine identifiability. Disentangling these is not directly possible from this matrix alone, but the per-class F1 scores reported in the confusion matrix (Figure 5.2.1) and the descriptive analysis presented in section 5.2.5 together provide a clearer view of which brand tiers carry distinguishable engagement and behavioural signatures.

5.2.4 Feature importance

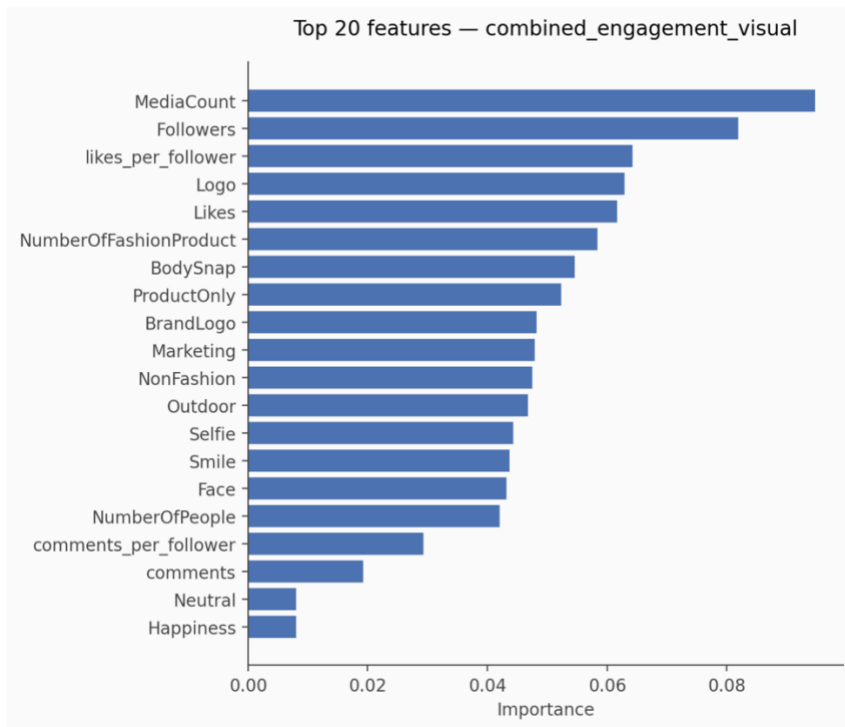


Figure 5.2.2 presents the feature matrix for combined engagement and visual

Figure 5.2.2 presents the top 20 features for the H2 test configuration. The three most important features are all engagement-based such as MediaCount with 0.094, Followers with 0.082, and Likes_per_follower with 0.064. Together with likes, the engagement features dominate the ranking. This explains why engagement alone outperformed visual features alone in the ablation which is that brand tier is more strongly signalled by who the user is than by what their photos look like.

Visual features appear from the 4th position (Logo) and onwards. The figure also shows how NumberOfFashionProduct, BodySnap, ProductOnly, BrandLogo, and Marketing are clustered between 0.04 and 0.06. This suggests that they are meaningful but secondary. The presence of Logo and BrandLogo near the top reflects the dataset's structure – Posts that are associated with luxury brands

The emotion scores sits at the bottom of the ranking Neutral and Happiness fall below 0.01, which suggest that facial-emotion probabilities contribute very little to the brand classification. The signal that distinguishes the brands lies in reach and engagement intensity, with visual content adding modest secondary information and facial emotion adding almost nothing

5.2.5 H3 descriptive evidence

Feature	Designer	High Street	Mega Couture	Small Couture
Selfie	0.038	0.045	0.032	0.019
BodySnap	0.320	0.389	0.163	0.282
ProductOnly	0.389	0.293	0.566	0.484
Face	0.341	0.416	0.235	0.274
Logo	0.509	0.503	0.672	0.406
Smile	0.100	0.123	0.077	0.068
Outdoor	0.130	0.159	0.063	0.085
NumberOfPeople	0.753	0.814	0.608	0.695
NumberOfFashionProduct	3.185	3.161	2.665	3.685

Table 5.2.3 presents the examples on mean visual and behavioural features values by brand category

Tabel 5.2.3 shows the mean visual and behavioural feature values for each brand category (Designer, High Street, Mega Couture and Small Couture). The table indicates

that High Street brands leads on people-centric content with highest rates on Selfie at 0.045, Bodysnap at 0.389, Face at 0.416 and on Smile, Outdoor and NumberOfPeople. Mega Couture shows the opposite med highest on ProductOnly at 0.566 and Logo at 0.672 with the lowest rate on Selfie, BodySnap and Face. Small Couture leads with the highest in NumberOfFashionProducts at 3.685 and one of the lowest selfie rates.

H3 is supported since the four categories exhibit systematically distinct posting behaviours. Table 5.2.3 shows how there is a people/selfie register for High Street brand, whereas Mega Couture exhibits a product/logo register. Since Small Couture has the highest rate on NumberOfFashionProducts it indicates that users post a more catalogue-style presentation of this brand. These differences exists descriptively, but as section 5.2.4 showed, they are insufficient to fully separate them in a classifier. The brands share enough behavioural overlap to confuse the model despite their measurable mean differences.

6 Discussion

Style Notes

In this chapter, the study will answer the research questions, analyze and assess the finding and discuss their implications. Lastly, a reference for future work will be provided.

6.1 Answering the research questions

The overall research question was:

How can computer vision and machine learning be applied to social media data to extract fashion attributes and classify consumer behavior across fashion brands categories on Instagram?

The study explored these aspects with three sub-questions:

RQ 1: What fashion attributes can be automatically extracted from images using YOLOv8, and what characterizes the visual content of consumer-generated fashion images?

The Visual Attribute Extraction (Pipeline 1) successfully extracts four structured attribute dimensions from raw Instagram images. First, the parent garment category via YOLOv8 trained on DeepFashion2. Second, the dominant colour via K-means clustering in CIELAB space, Third, the pattern complexity via Laplacian-variance. Lastly, the refined garment sub-category via CLIP zero-shot with high per-parent thresholds and prompt ensembling

Across the 397 input image, the pipeline was able to produce 601 garment-level attributes with (90.9% validated manually in section 5.1.5), and a calibrated 4.16% uncertain rate at the sub-category level. The visual content of the images is dominated by casual, upper-body clothing in muted neutral tones. The analysis revealed that long sleeve top, trousers and short sleeve top account for 84% of all detections, with the dominant colour distribution concentrated in navy, beige, cream and denim on around 55% of the garments. CLIP identifies sweatshirts, leggings, and blouses as the main sub-categories with the dominant parent categories. One could argue, that this characterization sketches a register of soft tailoring rather than formal fashion, which carries a visual distinction from commercial fashion marketing.

RQ 2: Can machine learning models classify Instagram content into fashion brand categories based on engagement and visual features, and what features drive this classification?

The Random Forest classifier achieves moderate but not a high performance on the H2 test.

The classifier achieves a macro-F1 of 0.619 using engagement features alone, and 0.591 when visual features are added. Classification is therefore possible, but the combination does not outperform engagement alone. The classification of Instagram content into fashion brand categories based on engagement and visual features is therefore possible. The feature-importance analysis shows that the classification is primarily driven by reach-related engagement features such as MediaCount, Followers, Likes_per_follower, and Likes which occupy the top four positions. Visual features also contribute meaningful information, but secondarily with Logo and ProductOnly being the most informative. Facial-emotion features were also studied but they contributed almost nothing to the results. Based on the different analysis, one could argue that, the classifier is reading brands mostly from who the user is based on their reach and posting frequency, rather than from what their content looks like. This is a substantive finding is that in this dataset, the visual content of a fashion post tells the model less about the brand tier than the post's engagement does.

RQ 3: How do visual attributes from images and consumer engagement patterns on Instagram complement each other in understanding fashion behavior on social media?

The two pipelines was able to provide complementary but partially conflicting views of consumer behaviour in fashion. Pipeline 1 describes what consumers wear and post which lends towards a casual, neutral, person-centric posts dominated by sweatshirts, leggings and blouses. Pipeline 2 describes how brand tiers related to specific categories differ in posting behaviour. It was revealed that High Street content is people-and-selfie heavy, Mega Couture content is product-and-logo heavy, Small Couture lends towards catalogue style and how Designer sits in between. These differences exist descriptively in the data (Section 5.2.5) and they constitute the strongest evidence for H3.

The conflict between the pipelines is analytically useful. Pipeline 1 finds consumer dressing and Pipeline 2 their behaviour. Both results can be true at the same time, and it was noted that content varies by brand tier. Neither of the two pipeline is sufficient to characterize consumer behaviour on Instagram, but together, they are able to describe two complementary dimensions of the same phenomenon.

Returning to the overarching questions on how computer vision and machine learning can be applied to social media data to extract fashion attributes and classify consumer behaviour across brand categories. The framework presented in this study offers a reproducible, two-pipeline answer. Computer vision can be applied to extract structured visual attributes from consumer-generated images at scale, which can be used to characterize what people wear. Machine learning, on the hand, can identify systematic behavioral differences across brands, which can be used to characterize how consumers engage with fashion brands. The two approaches are most informative when combined, since each pipeline answers questions, that the other cannot. The combination reveals patterns which is the gap between what consumers wear and how they posts within different brand categories, that information would not be visible from either modality alone.

6.2 Evaluating the hypotheses

H1: Visual Attribute Hypothesis: Partially supported

Visual fashion attributes extracted from images using automated computer vision techniques like YOLOv8 show distinct distributional patterns across garment types, colors, and patterns.

Garment categories and dominant colours show strongly distinct distributional patterns (as discussed in section 6.1). The pattern dimension provides weaker evidence, but for a methodological reason the quantile-bucket design produces an approximately uniform marginal distribution by construction. H1 is therefore supported on garment type and colour, methodologically inconclusive on pattern, and supported overall.

H2: Multimodal Superiority Hypothesis: Not supported

A machine learning model combining visual and engagement features outperforms single modality models in classifying Instagram content into fashion brand categories

The clean H2 test (engagement-only 0.619, visual-only 0.492, combined 0.591) shows that combining modalities does not outperform the best single modality. The rejection is itself substantive: the multimodal-superiority pattern reported in prior literature does not generalize to brand-tier classification on this dataset. The supplementary hashtag runs achieve much higher scores (0.94+), but it reflect brand-identity leakage from the dataset's hashtag-based collection method rather than recoverable signal.

H3: Consumer Behavior Hypothesis: Supported

Consumer-generated fashion content displays distinct engagement patterns, and these patterns vary systematically across fashion brand categories (High Street, Designer, Small Couture, Mega Couture).

The four brand categories exhibit systematically distinct posting registers (section 5.2.5). High Street people-centric, Mega Couture product-and-logo, Small Couture catalogue-style, Designer intermediate. These descriptive differences are not, however, sufficient to fully separate the tiers in a classifier, so H3 is supported at the level of aggregate posting register rather than per-post identifiability.

6.3 Relation to literature

The findings of this study agree, extend, and partially contradict with existing literature on data-driven fashion analysis from social media.

The study is consistent with Balloni et al. (2024), their system demonstrated how automated detection and extraction of garment and colours attributes from social media is possible. Pipeline 1 in this study replicates this core finding using YOLOv8 and CLIP rather than the

deep learning techniques used in Social4Fashion. During this supports the broader literature consensus that computer vision can reliably structure visual signals from images (Au et al., 2008; Jain, 2025; Choi et al., 2014). The use of engagement metrics as proxy for behaviour aligns the literature (Jain & Singh, 2025; Silvia et al., 2019). The descriptive evidence for H3 did confirm that engagement patterns vary across brand categories which is an empirical extension of arguments made in Balloni et al. (2024) and Ha et al. (2017).

This study extends the literature by operationalizing a fashion taxonomy of 13 parent categories and 53 refined categories, this approach extends the Social4Fashion framework, which classifies at a broad category level. Kim et al (2020) collected Instagram images as input to influencer profiling, where this study repurposes the same data to test classification. The study produced a validation framework for the image dataset which contributes a reproducible manual-validation tool for similar pipelines. Ha et al. (2017) constructed image-content classifiers (Selfie, BodySnap, Marketing, ProductOnly, Logo) trained on hand-labelled Instagram fashion data, and their classifier outputs accompany the metadata dataset used in this study. The features used in Pipeline 2 are therefore validated at the level of Ha et al.'s training corpus, but no per-post ground truth for these labels is available within the data used here. Pipeline 1, by contrast, processes raw images for which no prior ground truth exists, motivating the validation framework described in section 4.5.5.

6.4 Future research

This thesis has shown how several directions can extend naturally from the framework presented in this story.

The image-analyses was applied to a small sample of images collected between 2018 and 2019 from the Kim et al. (2020) family category, whereas future research could and should draw from a larger and more current dataset. This would allow descriptive findings about garment, colours and pattern distribution to tested against current trends, which has shifted in the current years. It could furthermore be interesting to use one dataset instead of two complementary datasets, this would enable direct multimodal modelling rather than the parallel design used here. In addition to that, a longitudinal dataset, capturing how engagement patterns and visual attributes evolve over time, would shift the analytical frame from descriptive characterization to a dynamic trend modelling. This would also allow test on trend-emergence patterns as discussion in section 2.5 (Barbiecore, Brat). For the attribute extraction it could be relevant to

use a different or extend pattern classifier, that can distinguish pattern types (stripes, floral, polka dots) rather than complexity (plain, subtle and patterned) to get a more precise analysis.

6.5 Implications

The framework developed in this study has practical implications for fashion brands, it demonstrates how it is possible to draw informative information about consumers through Instagram. The first pipeline analysis demonstrates how consumer-generated content namely from Instagram is able to extract meaningful attributes from images, while also filtering non-fashion related images from datasets. Brands can therefore apply this kind of analysis to identify which attributes circulate from consumer-generated content within their own markets, distinguishing the attributes to what consumers actually wear and post. The signal can inform stocking decisions, colour planning to align with marketing to the visual register of the brand's actual consumer base. The same logic extends to the other extracted attributes, that the study demonstrated in the classification results (section 5.2.5) which shows distinct posting registers across brand categories. The analysis revealed that High Streets users produce people-and-selfie content, Mega Couture content is centred around products and logo. Brands can use this kind of segmentation to align their own content strategy to match their consumer base.

The framework is, In this sense, not only a research tool but a method for understanding between brand-led and consumer-led visual narratives in fashion.

7 Conclusion

Closing the Digital Wardrobe

Fashion on social media evolves faster than the systems built to study it. The interpretable signals that consumers generate every day, such as the garments they wear, the colours they choose, the way they frame and engage with content are unstructured by default. They have therefore historically been read through intuition or sales data rather than through systematic analysis. This thesis set out to test whether these signals can instead be captured and analysed at scale through computer vision and machine learning to gain new insights.

The study addressed this challenge through two complementary pipelines. The Visual Attribute Extraction pipeline (1) applies YOLOv8, K-means colour clustering, Laplacian-variance pattern detection, and CLIP zero-shot refinement to consumer-generated images, producing a structured record of garment category, dominant colour, pattern complexity, and fine-grained sub-category. Pipeline 1 clearly and conclusively shows diverse and distinct distributional patterns across social media user's garments. The Consumer Behaviour Classification pipeline (2) applies a Random Forest classifier to Instagram metadata to identify and explain behavioural differences across brand categories. Together, the two pipelines describe two complementary dimensions of fashion behaviour namely what consumers wear, and how they present and engage with it.

The main findings are concrete. Pipeline 1 extracted 601 structured garment-level records from 397 consumer-generated images with 90.9% parent-category accuracy under manual validation, and revealed a visual register dominated by casual, neutral-toned, upper-body clothing. Pipeline 2 achieved a macro-F1 of 0.591 on the H2 test, with engagement features alone outperforming the combined model at 0.619, rejecting H2 and identifying reach-related signals (MediaCount, Followers, engagement rates) as the dominant predictors of brand tier. The behavioural analysis showed that the four brand categories exhibit distinct posting registers: High Street person-centric, Mega Couture product-and-logo, Small Couture catalogue-style, Designer intermediate supporting H3 at the descriptive level even where the classifier could not fully separate them.

The contribution of this study is a reproducible, two-pipeline framework rather than a definitive fashion-forecasting model. The framework demonstrates that structured, multi-dimensional

analysis of consumer fashion behaviour is feasible from publicly available social media data, using computer vision and machine learning components. The hierarchical fashion taxonomy of 13 parent categories and 53 fine-grained sub-labels, together with the manual validation framework, are methodological contributions that future research can extend and adapt.

The study's main limitations are inherent to the proof-of-concept design. Both datasets are cross-sectional snapshots from 2018–2019 and 2015 respectively, with no longitudinal dimension. The two datasets are also independent rather than image-and-metadata linked at the post level, which prevents direct multimodal modelling. The visual features used in Pipeline 2 are inherited from Ha et al. (2017) rather than independently validated, and causal interpretation of the behavioural patterns lies beyond the scope of a descriptive framework.

Returning to the question that opened this thesis: *“what can the data reveal about what people actually wear and how they express and engage with fashion online”*. The answer is that data can reveal a great deal, but the interpretation must remain careful. Consumer-generated fashion content carries a visual register distinct from commercial fashion communication, and brand-tier behavioural patterns are systematic enough to be detected by automated methods. Yet the same analysis also shows that engagement is shaped by reach as much as by content, that fine-grained classification remains hard, and that the strongest signals in a dataset can sometimes reveal more about how the data was collected than about the consumers whose behaviour it describes. Social media fashion data can be systematized, but the conclusions it supports are bounded by the structure of the data itself. The framework presented contributes to the constantly evolving corpus of fashion analysis, to help improve and further its effectiveness in understanding consumer behaviour.

8 Bibliography

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Appendix

Deklarationsformular AI

Description on Fashiondata from Ha et al.